

Red Hat OpenShift Data Foundation 4.22

Scaling storage

Instructions for scaling operations in OpenShift Data Foundation

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Instructions for scaling operations in OpenShift Data Foundation

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Abstract

This document explains scaling options for Red Hat OpenShift Data Foundation.

Table of Contents

PREFACE 4

PROVIDING FEEDBACK ON RED HAT DOCUMENTATION 5

CHAPTER 1. INTRODUCTION TO SCALING STORAGE 6
1.1. SUPPORTED DEPLOYMENTS FOR RED HAT OPENSHIFT DATA FOUNDATION 6

CHAPTER 2. REQUIREMENTS FOR SCALING STORAGE 8

CHAPTER 3. SCALING STORAGE CAPACITY OF AWS OPENSHIFT DATA FOUNDATION CLUSTER 9
3.1. SCALING UP STORAGE CAPACITY ON A CLUSTER 9
3.1.1. Scaling up storage capacity on an AWS cluster by resizing existing OSDs 11
3.2. SCALING OUT STORAGE CAPACITY ON A AWS CLUSTER 12
3.2.1. Adding a node 13
3.2.1.1. Adding a node to an installer-provisioned infrastructure 13
3.2.1.2. Adding a node to an user-provisioned infrastructure 14
3.2.2. Scaling up storage capacity 15
3.3. ENABLING AUTOMATIC CAPACITY SCALING 15
3.4. SCALING NON-RESILIENT POOLS 16

CHAPTER 4. SCALING STORAGE OF BARE METAL OPENSHIFT DATA FOUNDATION CLUSTER 18
4.1. SCALING UP A CLUSTER CREATED USING LOCAL STORAGE DEVICES 18
4.1.1. Scaling up storage capacity on a cluster by resizing existing OSDs 20
4.2. SCALING OUT STORAGE CAPACITY ON A BARE METAL CLUSTER 21
4.2.1. Adding a node 22
4.2.1.1. Adding a node to an installer-provisioned infrastructure 22
4.2.1.2. Adding a node using a local storage device 23
4.2.2. Scaling up storage capacity 25

CHAPTER 5. SCALING STORAGE USING MULTIPLE DEVICE CLASS IN THE SAME CLUSTER FOR LOCAL STORAGE DEPLOYMENTS 26
5.1. CREATING A NEW LOCALVOLUMESET 26
5.2. ATTACHING STORAGE FOR A NEW DEVICE SET 28
5.3. ATTACHING STORAGE FOR A NEW DEVICE SET BY USING DISKS OF SAME TYPE ON THE SAME NODES 31

CHAPTER 6. SCALING STORAGE OF VMWARE OPENSHIFT DATA FOUNDATION CLUSTER 35
6.1. SCALING UP STORAGE ON A DYNAMICALLY PROVISIONED VMWARE CLUSTER 35
6.2. SCALING UP A CLUSTER CREATED USING LOCAL STORAGE DEVICES 37
6.2.1. Scaling up storage capacity on a cluster by resizing existing OSDs 39
6.3. ENABLING AUTOMATIC CAPACITY SCALING 40
6.4. SCALING OUT STORAGE CAPACITY ON A VMWARE CLUSTER 41
6.4.1. Adding a node to an installer-provisioned infrastructure 41
6.4.2. Adding a node to an user-provisioned infrastructure 42
6.4.3. Adding a node using a local storage device 43
6.4.4. Scaling up storage capacity 45
6.5. SCALING NON-RESILIENT POOLS 46

CHAPTER 7. SCALING STORAGE OF MICROSOFT AZURE OPENSHIFT DATA FOUNDATION CLUSTER 48
7.1. SCALING UP STORAGE CAPACITY ON A CLUSTER 48
7.1.1. Scaling up storage capacity on a cluster by resizing existing OSDs 50
7.2. ENABLING AUTOMATIC CAPACITY SCALING 51
7.3. SCALING OUT STORAGE CAPACITY ON A MICROSOFT AZURE CLUSTER 52
7.3.1. Adding a node to an installer-provisioned infrastructure 52

7.3.2. Scaling up storage capacity 53
7.4. SCALING NON-RESILIENT POOLS 53

CHAPTER 8. SCALING STORAGE CAPACITY OF GCP OPENSHIFT DATA FOUNDATION CLUSTER 55
8.1. SCALING UP STORAGE CAPACITY ON A CLUSTER 55
8.1.1. Scaling up storage capacity on a cluster by resizing existing OSDs 57
8.2. ENABLING AUTOMATIC CAPACITY SCALING 58
8.3. SCALING OUT STORAGE CAPACITY ON A GCP CLUSTER 59
8.3.1. Adding a node 59
8.3.1.1. Adding a node to an installer-provisioned infrastructure 59
8.3.2. Scaling up storage capacity 60
8.4. SCALING NON-RESILIENT POOLS 60

CHAPTER 9. SCALING STORAGE OF IBM Z OR IBM LINUXONE OPENSHIFT DATA FOUNDATION CLUSTER 63
9.1. SCALING UP STORAGE BY ADDING CAPACITY TO YOUR OPENSHIFT DATA FOUNDATION NODES ON IBM Z OR IBM LINUXONE INFRASTRUCTURE 63
9.2. SCALING OUT STORAGE CAPACITY ON A IBM Z OR IBM LINUXONE CLUSTER 66
9.2.1. Adding a node using a local storage device 66
9.2.2. Scaling up storage capacity 68

CHAPTER 10. SCALING STORAGE OF IBM POWER OPENSHIFT DATA FOUNDATION CLUSTER 69
10.1. SCALING UP STORAGE BY ADDING CAPACITY TO YOUR OPENSHIFT DATA FOUNDATION NODES ON IBM POWER INFRASTRUCTURE USING LOCAL STORAGE DEVICES 69
10.2. SCALING OUT STORAGE CAPACITY ON A IBM POWER CLUSTER 72
10.2.1. Adding a node using a local storage device on IBM Power 73
10.2.2. Scaling up storage capacity 74

CHAPTER 11. SCALING STORAGE CAPACITY OF IBM FLASHSYSTEM CLUSTER 75
11.1. SCALING UP STORAGE CAPACITY ON A CLUSTER 75
11.2. SCALING OUT STORAGE CAPACITY ON AN IBM FLASHSYSTEM CLUSTER 77
11.2.1. Adding a node 77
11.2.1.1. Adding a node to an installer-provisioned infrastructure 77
11.2.1.2. Adding a node to an user-provisioned infrastructure 78
11.2.2. Scaling up storage capacity 79

PREFACE

This document describes how to scale the storage capacity of your OpenShift Data Foundation cluster on various platforms.

PROVIDING FEEDBACK ON RED HAT DOCUMENTATION

We appreciate your input on our documentation. Do let us know how we can make it better.

To give feedback, create a Jira ticket:

Procedure

1. Log in to the Jira.
2. Click **Create** in the top navigation bar
3. Enter a descriptive title in the Summary field.
4. Enter your suggestion for improvement in the Description field. Include links to the relevant parts of the documentation.
5. Select **Documentation** in the Components field.
6. Click Create at the bottom of the dialogue.

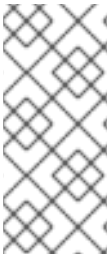
CHAPTER 1. INTRODUCTION TO SCALING STORAGE

Red Hat OpenShift Data Foundation is a highly scalable storage system. OpenShift Data Foundation allows you to scale by adding the disks in the multiple of three, or three or any number depending upon the deployment type.

- For internal (dynamic provisioning) deployment mode, you can increase the capacity by adding 3 disks at a time.
- For internal-attached (Local Storage Operator based) mode, you can deploy with less than 3 failure domains.

With flexible scale deployment enabled, you can scale up by adding any number of disks. For deployment with 3 failure domains, you will be able to scale up by adding disks in the multiple of 3.

For scaling your storage in external mode, see the Red Hat Ceph Storage documentation.



NOTE

You can use a maximum of twelve storage devices per node. The high number of storage devices will lead to a higher recovery time during the loss of a node. This recommendation ensures that nodes stay below the cloud provider dynamic storage device attachment limits, and limits the recovery time after node failure with local storage devices.

While scaling, you must ensure that there are enough CPU and Memory resources as per scaling requirement.

Supported storage classes by default

- **gp2-csi** on AWS
- **thin** on VMware
- **managed-csi** on Microsoft Azure, including enabled performance plus

1.1. SUPPORTED DEPLOYMENTS FOR RED HAT OPENSHIFT DATA FOUNDATION

- User-provisioned infrastructure:
  - Amazon Web Services (AWS)
  - VMware
  - Bare metal
  - IBM Power
  - IBM Z or IBM® LinuxONE
- Installer-provisioned infrastructure:
  - Amazon Web Services (AWS)

- Microsoft Azure
- VMware
- Bare metal

Additional resources

- Red Hat Ceph Storage documentation

CHAPTER 2. REQUIREMENTS FOR SCALING STORAGE

Before you proceed to scale the storage nodes, refer to the following sections to understand the node requirements for your specific Red Hat OpenShift Data Foundation instance:

- Platform requirements
- Resource requirements
- Storage device requirements
  - Dynamic storage devices
  - Local storage devices
  - Capacity planning



IMPORTANT

Always ensure that you have plenty of storage capacity.

If storage ever fills completely, it is not possible to add capacity or delete or migrate content away from the storage to free up space completely. Full storage is very difficult to recover.

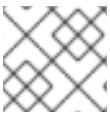
Capacity alerts are issued when cluster storage capacity reaches 75% (near-full) and 85% (full) of total capacity. Always address capacity warnings promptly, and review your storage regularly to ensure that you do not run out of storage space.

If storage capacity reaches 85% full state, Ceph may report **HEALTH_ERR** and prevent IO operations. In this case, you can increase the **full** ratio temporarily so that cluster rebalance can take place. For steps to increase the **full** ratio, see Setting Ceph OSD full thresholds using the ODF CLI tool.

If you do run out of storage space completely, contact Red Hat Customer Support.

CHAPTER 3. SCALING STORAGE CAPACITY OF AWS OPENSHIFT DATA FOUNDATION CLUSTER

To scale the storage capacity of your configured Red Hat OpenShift Data Foundation worker nodes on AWS cluster, you can increase the capacity by adding three disks at a time. Three disks are needed since OpenShift Data Foundation uses a replica count of 3 to maintain the high availability. So the amount of storage consumed is three times the usable space.



NOTE

Usable space might vary when encryption is enabled or replica 2 pools are being used.

3.1. SCALING UP STORAGE CAPACITY ON A CLUSTER

To increase the storage capacity in a dynamically created storage cluster on an user-provisioned infrastructure, you can add storage capacity and performance to your configured Red Hat OpenShift Data Foundation worker nodes.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.
- The disk should be of the same size and type as used during initial deployment.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Storage → Storage cluster**.
    a. Click **Actions** at the top right of the page to extend the options menu.
    b. Select **Add Capacity** from the options menu.
    c. Select the **Storage Class**. Choose the storage class which you wish to use to provision new storage devices.
    d. Click **Add**.
3. To check the status, navigate to **Storage → Storage cluster** and verify that the **Storage Cluster** in the Status card has a green tick.

Verification

- Verify the **Raw Capacity** card.
    a. In the OpenShift Web Console, click **Storage → Storage cluster**.
    b. In the **Block and File** tab, check the **Raw Capacity** card.
    Note that the capacity increases based on your selections.



NOTE

The raw capacity does not take replication into account and shows the full capacity.

- Verify that the new object storage devices (OSDs) and their corresponding new Persistent Volume Claims (PVCs) are created.
  - To view the state of the newly created OSDs:
    a. Click **Workloads → Pods** from the OpenShift Web Console.
    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

  - To view the state of the PVCs:
    a. Click **Storage → Persistent Volume Claims** from the OpenShift Web Console.
    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

- Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSD devices are encrypted.
  a. Identify the nodes where the new OSD pods are running.

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
```

<OSD-pod-name>

Is the name of the OSD pod.
For example:

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
```

Example output:

```
NODE
compute-1
```

  b. For each of the nodes identified in the previous step, do the following:
    i. Create a debug pod and open a chroot environment for the selected hosts.

```
$ oc debug node/<node-name>
```

<node-name>

Is the name of the node.

```
$ chroot /host
```

ii. Check for the **crypt** keyword beside the **ocs-deviceset** names.

```
$ lsblk
```

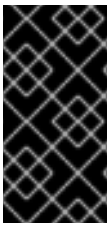


IMPORTANT

Cluster reduction is supported only with the Red Hat Support Team's assistance.

3.1.1. Scaling up storage capacity on an AWS cluster by resizing existing OSDs

To increase the storage capacity on a cluster, you can add storage capacity by resizing existing OSDs.



IMPORTANT

Resizing existing OSDs can only be done once every 6 hours due to AWS limitations. If you exceed this amount within the 6 hour time frame, you will receive a warning that includes the note **You've reached the maximum modification rate per volume limit. Wait at least 6 hours between modifications per EBS volume.**

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Update the **dataPVCTemplate** size for the **storageDeviceSets** in the StorageCluster CR with the new desired size. For example:

```
apiVersion: ocs.openshift.io/v1
kind: StorageCluster
metadata:
  namespace: openshift-storage
  name: example-storagecluster
spec:
  manageNodes: false
  monPVCTemplate:
    spec:
      storageClassName:
      accessModes:
      - ReadWriteOnce
      resources:
        requests:
          storage: 10Gi
```

```
storageDeviceSets:
- name: example-deviceset
  count: 3
  resources: {}
  placement: {}
  dataPVCTemplate:
    spec:
      storageClassName:
      accessModes:
      - ReadWriteOnce
      volumeMode: Block
      resources:
        requests:
          storage: 1Ti
  portable: true
```

In this example YAML, the **storage** parameter under **storageDeviceSets** reflects the size change of **1Ti**.

2. Wait for the OSDs to restart.

3. Confirm that the resize took effect:

```
$ oc get pvc -l ceph.rook.io/DeviceSet
```

Example output:

```
% oc get pvc -l ceph.rook.io/DeviceSet
NAME                STATUS   VOLUME                                     CAPACITY   ACCESS MODES
STORAGECLASS                  AGE
rook-ceph-osd-a4681   Bound    pvc-c3868d38-6bd9-4c8d-91e1-a4681e4cb8c1   1Ti
RWO            gp2-csi   176m
```

Verify the resize is complete in the **CAPACITY** column.

4. If the resize did not take effect, restart the OSD pods again. It may take multiple restarts for the resize to complete.

3.2. SCALING OUT STORAGE CAPACITY ON A AWS CLUSTER

OpenShift Data Foundation is highly scalable. It can be scaled out by adding new nodes with required storage and enough hardware resources in terms of CPU and RAM. Practically there is no limit on the number of nodes which can be added but from the support perspective 2000 nodes is the limit for OpenShift Data Foundation.

Scaling out storage capacity can be broken down into two steps

- Adding new node
- Scaling up the storage capacity



NOTE

OpenShift Data Foundation does not support heterogeneous OSD/Disk sizes.

3.2.1. Adding a node

You can add nodes to increase the storage capacity when existing worker nodes are already running at their maximum supported OSDs or there are not enough resources to add new OSDs on the existing nodes. It is always recommended to add nodes in the multiple of three, each of them in different failure domains.

While it is recommended to add nodes in the multiple of three, you still have the flexibility to add one node at a time in the flexible scaling deployment. Refer to the Knowledgebase article Verify if flexible scaling is enabled.



NOTE

OpenShift Data Foundation does not support heterogeneous disk size and types. The new nodes to be added should have the disk of the same type and size which was used during OpenShift Data Foundation deployment.

3.2.1.1. Adding a node to an installer-provisioned infrastructure

Use this procedure to add a new node to an installer-provisioned infrastructure cluster.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Navigate to **Compute → Machine Sets**.
2. On the machine set where you want to add nodes, select **Edit Machine Count**
   a. Add the amount of nodes, and click **Save**.
   b. Click **Compute → Nodes** and confirm if the new node is in **Ready** state.
3. Apply the OpenShift Data Foundation label to the new node.
   a. For the new node, click **Action menu (⋮) → Edit Labels**.
   b. Add **cluster.ocs.openshift.io/openshift-storage**, and click **Save**.



NOTE

It is recommended to add 3 nodes, one each in different zones. You must add 3 nodes and perform this procedure for all of them. In case of bare metal installer-provisioned infrastructure deployment, you must expand the cluster first. For instructions, see Expanding the cluster.

Verification

1. Execute the following command in the terminal and verify that the new node is present in the output:

```
$ oc get nodes --show-labels | grep cluster.ocs.openshift.io/openshift-storage= |cut -d' ' -f1
```

2. On the OpenShift web console, click **Workloads → Pods**, confirm that at least the following pods on the new node are in **Running** state:
   - **openshift-storage.cephfs.csi.ceph.com-***
   - **openshift-storage.rbd.csi.ceph.com-***

3.2.1.2. Adding a node to an user-provisioned infrastructure

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Depending on the type of infrastructure, perform the following steps:
   a. Get a new machine with the required infrastructure. See Platform requirements.
   b. Create a new OpenShift Container Platform worker node using the new machine.
2. Check for certificate signing requests (CSRs) that are in **Pending** state.

```
$ oc get csr
```

3. Approve all the required CSRs for the new node.

```
$ oc adm certificate approve <Certificate_Name>
```

<Certificate_Name>
Is the name of the CSR.

4. Click **Compute → Nodes**, confirm if the new node is in **Ready** state.
5. Apply the OpenShift Data Foundation label to the new node using any one of the following:

From User interface

a. For the new node, click **Action Menu (⋮) → Edit Labels**.
b. Add **cluster.ocs.openshift.io/openshift-storage**, and click **Save**.

From Command line interface

- Apply the OpenShift Data Foundation label to the new node.

```
$ oc label node <new_node_name> cluster.ocs.openshift.io/openshift-storage=""
```

<new_node_name>
Is the name of the new node.

Verification

1. Execute the following command in the terminal and verify that the new node is present in the output:

```
$ oc get nodes --show-labels | grep cluster.ocs.openshift.io/openshift-storage= |cut -d' ' -f1
```

2. On the OpenShift web console, click **Workloads → Pods**, confirm that at least the following pods on the new node are in **Running** state:
   - **openshift-storage.cephfs.csi.ceph.com-***
   - **openshift-storage.rbd.csi.ceph.com-***

3.2.2. Scaling up storage capacity

To scale up storage capacity, see Scaling up storage capacity on a cluster .

3.3. ENABLING AUTOMATIC CAPACITY SCALING

You can enable automatic capacity scaling on clusters deployed using dynamic storage devices. When automatic capacity scaling is enabled, additional raw capacity equivalent to the configured deployment size is automatically added to the cluster when used capacity reaches 70%. This ensures your deployment scales seamlessly to meet demand.

This option is disabled in lean profile mode, LSO deployment, and external mode deployment.



IMPORTANT

This may incur additional costs for the underlying storage.

Prerequisites

- Administrative privilege to the OpenShift Container Platform Console.
- A running OpenShift Data Foundation Storage Cluster.



IMPORTANT

IBM cloud **10iops-tier** workloads are not supported.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Storage → Storage cluster**.
3. Click **Actions** at the top right of the page to extend the options menu.
4. Select **Automatic capacity scaling** from the options menu.
5. In the **Automatic capacity scaling** page, select the **Enable automatic capacity scaling for your cluster** checkbox.

6. Set the cluster expansion limit from the dropdown. This is the maximum the cluster can expand in the cloud. Automatic scaling is suspended if this limit is exceeded.

7. Click **Save changes**.

3.4. SCALING NON-RESILIENT POOLS

After enabling the single replica feature, you can increase the number of non-resilient pools by updating the pool count in the **StorageCluster** custom resource. Each additional pool is created in a separate failure domain and requires one new storage device per failure domain.



WARNING

Non-resilient pools have no data replication. Increasing the pool count does not provide data redundancy. Ensure that your applications manage their own replication before scaling.

Prerequisites

- The single replica feature is already enabled. For more information, see Storage class with single replica.
- For LSO-based deployments, ensure that the required number of unused local storage devices are correctly labeled on the nodes before scaling.

Procedure

1. Run the following command to increase the non-resilient pool count:

```
$ oc patch storagecluster ocs-storagecluster \
  -n openshift-storage \
  --type=json \
  --patch='[
    {
      "op": "replace",
      "path": "/spec/managedResources/cephNonResilientPools/count",
      "value": 2
    }
  ]'
```

Replace **2** with the desired number of non-resilient pools. The value must not exceed the number of available failure domains.



NOTE

For LSO-based deployments, the operator creates one OSD per local device per failure domain. Ensure that the unused local devices are available on the nodes before running this command.

2. Verify that the **StorageCluster** is in **Ready** state:

```
$ oc get storagecluster -n openshift-storage
```

Example output:

```
NAME                 AGE   PHASE   EXTERNAL   CREATED AT             VERSION
ocs-storagecluster   10m   Ready              2024-02-05T13:56:15Z   4.22.0
```

3. Verify that the new **CephBlockPools** are created for each failure domain and are in **Ready** state:

```
$ oc get cephblockpools -n openshift-storage
```

Example output:

```
NAME                                      PHASE
ocs-storagecluster-cephblockpool          Ready
ocs-storagecluster-cephblockpool-us-east-1a   Ready
ocs-storagecluster-cephblockpool-us-east-1b   Ready
ocs-storagecluster-cephblockpool-us-east-1c   Ready
```

4. Verify that the new OSD pods are in **Running** state:

```
$ oc get pods -n openshift-storage | grep osd
```

CHAPTER 4. SCALING STORAGE OF BARE METAL OPENSHIFT DATA FOUNDATION CLUSTER

To scale the storage capacity of your configured Red Hat OpenShift Data Foundation worker nodes on your bare metal cluster, you can increase the capacity by adding three disks at a time. Three disks are needed since OpenShift Data Foundation uses a replica count of 3 to maintain the high availability. So the amount of storage consumed is three times the usable space.



NOTE

Usable space might vary when encryption is enabled or replica 2 pools are being used.

4.1. SCALING UP A CLUSTER CREATED USING LOCAL STORAGE DEVICES

To scale up an OpenShift Data Foundation cluster which was created using local storage devices, you need to add a new disk to the storage node. The new disks size must be of the same size as the disks used during the deployment because OpenShift Data Foundation does not support heterogeneous disks/OSDs.

For deployments having three failure domains, you can scale up the storage by adding disks in the multiples of three, with the same number of disks coming from nodes in each of the failure domains. For example, if we scale by adding six disks, two disks are taken from nodes in each of the three failure domains. If the number of disks is not in multiples of three, it will only consume the disk to the maximum in the multiple of three while the remaining disks remain unused.

For deployments having less than three failure domains, there is a flexibility to add any number of disks. Make sure to verify that flexible scaling is enabled. For information, refer to the Knowledgebase article Verify if flexible scaling is enabled .



NOTE

Flexible scaling features get enabled at the time of deployment and cannot be enabled or disabled later on.

Prerequisites

- Administrative privilege to the OpenShift Container Platform Console.
- A running OpenShift Data Foundation Storage Cluster.
- Make sure that the disks to be used for scaling are attached to the storage node
- Make sure that **LocalVolumeDiscovery** and **LocalVolumeSet** objects are created.

To add capacity, you can either use a storage class that you provisioned during the deployment or any other storage class that matches the filter.

Procedure

1. In the OpenShift Web Console, click **Storage → Storage cluster**.
    a. Click **Actions** at the top right of the page to extend the options menu.

b. Select **Add Capacity** from the options menu.

c. Select the **Storage Class** for which you added disks or the new storage class depending on your requirement. Available Capacity displayed is based on the local disks available in storage class.

d. Click **Add**.

2. To check the status, navigate to **Storage → Storage cluster** and verify that the **Storage Cluster** in the **Status** card has a green tick.

Verification

- Verify the **Raw Capacity** card.

  a. In the OpenShift Web Console, click **Storage → Storage cluster**.

  b. In the **Block and File** tab, check the **Raw Capacity** card.
  Note that the capacity increases based on your selections.



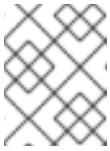
NOTE

The raw capacity does not take replication into account and shows the full capacity.

- Verify that the new OSDs and their corresponding new Persistent Volume Claims (PVCs) are created.
  - To view the state of the newly created OSDs:

    a. Click **Workloads → Pods** from the OpenShift Web Console.

    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

  - To view the state of the PVCs:

    a. Click **Storage → Persistent Volume Claims** from the OpenShift Web Console.

    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

- Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSD devices are encrypted.

  a. Identify the nodes where the new OSD pods are running.

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
```

<OSD-pod-name>

Is the name of the OSD pod.
For example:

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
```

Example output:

```
NODE
compute-1
```

b. For each of the nodes identified in the previous step, do the following:

i. Create a debug pod and open a chroot environment for the selected host(s).

```
$ oc debug node/<node-name>
```

<node-name>

Is the name of the node.

```
$ chroot /host
```

ii. Check for the **crypt** keyword beside the **ocs-deviceset** names.

```
$ lsblk
```



IMPORTANT

Cluster reduction is supported only with the Red Hat Support Team's assistance.

4.1.1. Scaling up storage capacity on a cluster by resizing existing OSDs

To increase the storage capacity on a cluster, you can add storage capacity by resizing existing OSDs.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Update the **dataPVCTemplate** size for the **storageDeviceSets** in the StorageCluster CR with the new desired size. For example:

```
apiVersion: ocs.openshift.io/v1
```

```
kind: StorageCluster
metadata:
  namespace: openshift-storage
  name: example-storagecluster
spec:
  manageNodes: false
  monPVCTemplate:
    spec:
      storageClassName:
      accessModes:
      - ReadWriteOnce
      resources:
        requests:
          storage: 10Gi
  storageDeviceSets:
  - name: example-deviceset
    count: 3
    resources: {}
    placement: {}
    dataPVCTemplate:
      spec:
        storageClassName:
        accessModes:
        - ReadWriteOnce
        volumeMode: Block
        resources:
          requests:
            storage: 1Ti
    portable: true
```

In this example YAML, the **storage** parameter under **storageDeviceSets** reflects the size change of **1Ti**.

2. Wait for the OSDs to restart.

3. Confirm that the resize took effect:

```
$ oc get pvc -l ceph.rook.io/DeviceSet
```

Example output:

```
% oc get pvc -l ceph.rook.io/DeviceSet
NAME                STATUS   VOLUME                                     CAPACITY   ACCESS MODES
STORAGECLASS                  AGE
rook-ceph-osd-a4681   Bound    pvc-c3868d38-6bd9-4c8d-91e1-a4681e4cb8c1   1Ti
RWO            gp2-csi   176m
```

Verify the resize is complete in the **CAPACITY** column.

4. If the resize did not take effect, restart the OSD pods again. It may take multiple restarts for the resize to complete.

4.2. SCALING OUT STORAGE CAPACITY ON A BARE METAL CLUSTER

OpenShift Data Foundation is highly scalable. It can be scaled out by adding new nodes with required

storage and enough hardware resources in terms of CPU and RAM. There is no limit on the number of nodes which can be added. However, from the technical support perspective, 2000 nodes is the limit for OpenShift Data Foundation.

Scaling out storage capacity can be broken down into two steps

- Adding new node
- Scaling up the storage capacity



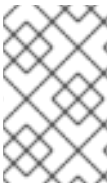
NOTE

OpenShift Data Foundation does not support heterogeneous OSD/Disk sizes.

4.2.1. Adding a node

You can add nodes to increase the storage capacity when existing worker nodes are already running at their maximum supported OSDs or there are not enough resources to add new OSDs on the existing nodes. It is always recommended to add nodes in the multiple of three, each of them in different failure domains.

While it is recommended to add nodes in the multiple of three, you still have the flexibility to add one node at a time in the flexible scaling deployment. Refer to the Knowledgebase article Verify if flexible scaling is enabled.



NOTE

OpenShift Data Foundation does not support heterogeneous disk size and types. The new nodes to be added should have the disk of the same type and size which was used during OpenShift Data Foundation deployment.

4.2.1.1. Adding a node to an installer-provisioned infrastructure

Use this procedure to add a new node to an installer-provisioned infrastructure cluster.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Navigate to **Compute → Machine Sets**.
2. On the machine set where you want to add nodes, select **Edit Machine Count**
   a. Add the amount of nodes, and click **Save**.
   b. Click **Compute → Nodes** and confirm if the new node is in **Ready** state.
3. Apply the OpenShift Data Foundation label to the new node.
   a. For the new node, click **Action menu (⋮) → Edit Labels**.

b. Add **cluster.ocs.openshift.io/openshift-storage**, and click **Save**.



NOTE

It is recommended to add 3 nodes, one each in different zones. You must add 3 nodes and perform this procedure for all of them. In case of bare metal installer-provisioned infrastructure deployment, you must expand the cluster first. For instructions, see Expanding the cluster.

Verification

1. Execute the following command in the terminal and verify that the new node is present in the output:

```
$ oc get nodes --show-labels | grep cluster.ocs.openshift.io/openshift-storage= |cut -d' ' -f1
```

2. On the OpenShift web console, click **Workloads → Pods**, confirm that at least the following pods on the new node are in **Running** state:

   - **openshift-storage.cephfs.csi.ceph.com-***
   - **openshift-storage.rbd.csi.ceph.com-***

4.2.1.2. Adding a node using a local storage device

You can add nodes to increase the storage capacity when existing worker nodes are already running at their maximum supported OSDs or when there are not enough resources to add new OSDs on the existing nodes.

Add nodes in the multiple of 3, each of them in different failure domains. Though it is recommended to add nodes in multiples of 3 nodes, you have the flexibility to add one node at a time in flexible scaling deployment. See Knowledgebase article Verify if flexible scaling is enabled



NOTE

OpenShift Data Foundation does not support heterogeneous disk size and types. The new nodes to be added should have the disk of the same type and size which was used during initial OpenShift Data Foundation deployment.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Depending on the type of infrastructure, perform the following steps:

   a. Get a new machine with the required infrastructure. See Platform requirements.

   b. Create a new OpenShift Container Platform worker node using the new machine.

2. Check for certificate signing requests (CSRs) that are in **Pending** state.

```
$ oc get csr
```

3. Approve all the required CSRs for the new node.

```
$ oc adm certificate approve <Certificate_Name>
```

<Certificate_Name>

Is the name of the CSR.

4. Click **Compute → Nodes**, confirm if the new node is in **Ready** state.

5. Apply the OpenShift Data Foundation label to the new node using any one of the following:

From User interface

a. For the new node, click **Action Menu (⋮) → Edit Labels**.

b. Add **cluster.ocs.openshift.io/openshift-storage**, and click **Save**.

From Command line interface

- Apply the OpenShift Data Foundation label to the new node.

```
$ oc label node <new_node_name> cluster.ocs.openshift.io/openshift-storage=""
```

<new_node_name>

Is the name of the new node.

6. Click **Ecosystem → Installed Operators** from the OpenShift Web Console.
From the **Project** drop-down list, make sure to select the project where the Local Storage Operator is installed.

7. Click **Local Storage**.

8. Click the **Local Volume Discovery** tab.

a. Beside the **LocalVolumeDiscovery**, click Action menu **(⋮) → Edit Local Volume Discovery**.

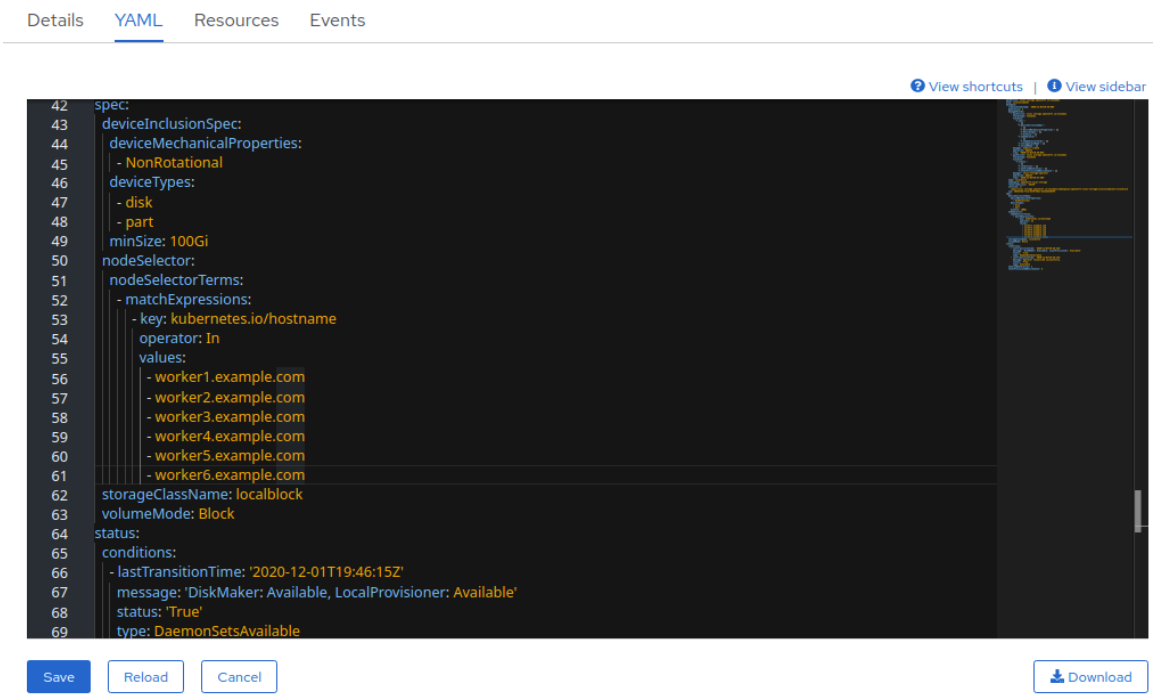
b. In the YAML, add the hostname of the new node in the **values** field under the node selector.

c. Click **Save**.

9. Click the **Local Volume Sets** tab.

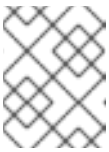
a. Beside the **LocalVolumeSet**, click Action menu **(⋮) → Edit Local Volume Set**

b. In the YAML, add the hostname of the new node in the **values** field under the **node selector**.



YAML showing the addition of new hostnames

c. Click **Save**.



NOTE

It is recommended to add 3 nodes, one each in different zones. You must add 3 nodes and perform this procedure for all of them.

Verification

1. Execute the following command in the terminal and verify that the new node is present in the output:

```
$ oc get nodes --show-labels | grep cluster.ocs.openshift.io/openshift-storage= |cut -d' ' -f1
```

2. On the OpenShift web console, click **Workloads → Pods**, confirm that at least the following pods on the new node are in **Running** state:

- **openshift-storage.cephfs.csi.ceph.com-***
- **openshift-storage.rbd.csi.ceph.com-***

4.2.2. Scaling up storage capacity

To scale up storage capacity, see Scaling up storage by adding capacity .

CHAPTER 5. SCALING STORAGE USING MULTIPLE DEVICE CLASS IN THE SAME CLUSTER FOR LOCAL STORAGE DEPLOYMENTS

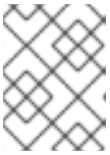
OpenShift Data Foundation supports creating multiple device classes for OSDs within the same cluster. Defining additional device classes provides flexibility in how storage devices are organized and allows you to:

- Use different types of disks on the same nodes
- Use disks of the same type on the same nodes
- Use different-sized disks of the same type on the same or different nodes
- Isolate disks of the same type across different sets of nodes
- Combine different resource types, such as local disks and SAN-based LUNs

Overview of the required tasks

To scale storage by using multiple device classes, complete the following tasks:

1. Prepare the disks.
   Attach new disks on the same or additional nodes and ensure they can be uniquely identified by a LocalVolumeSet.

   NOTE

   Update the **maxSize** or **disksFilter** parameter of the existing LocalVolumeSet (such as **localblock**) to prevent it from claiming the new PVs.

2. Create a new LocalVolumeSet.
   Define a LocalVolumeSet resource that represents the new device class.

3. Attach new storage.
   Attach the storage created by the new LocalVolumeSet so that it can be consumed by the cluster.

5.1. CREATING A NEW LOCALVOLUMESET

Use this procedure when you want to use devices of the same type but with different sizes.

Prerequisites

- Update the **maxSize** parameter of the existing LocalVolumeSet (for example, **localblock**) to ensure that it does not claim the newly created persistent volumes (PVs).

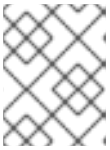
```
$ oc -n openshift-local-storage patch localvolumesets.local.storage.openshift.io localblock \
  -n openshift-local-storage \
  -p '{"spec": {"deviceInclusionSpec": {"maxSize": "120Gi"}}}' \
  --type merge
```

In this example, the existing LocalVolumeSet created during deployment might not have a **maxSize** value. Setting the limit to **120Gi** ensures that new disks with a higher size (for example, **130Gi**) are claimed only by the new LocalVolumeSet and do not overlap with the existing one.

- When creating the new LocalVolumeSet, plan a unique filter to identify the intended disks. You can differentiate disks by node, disk size, or disk type.
- Add the new disks. For example, add three new SSD or NVMe disks sized **130Gi**.

Procedure

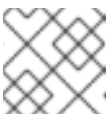
1. In the OpenShift Web Console, click **Ecosystem → Installed Operators**.
2. From the **Project** drop-down list, select the project where the **Local Storage Operator** is installed.
3. Click **Local Storage**.
4. Click the **Local Volume Sets** tab.
5. On the **Local Volume Sets** page, click **Create Local Volume Set**
6. Enter a name for the **LocalVolumeSet** and the associated **StorageClass**.



NOTE

The storage class name defaults to the LocalVolumeSet name, but it can be modified.

7. Under **Filter Disks By**, select one of the following options:
   - **Disks on all nodes**: Uses all matching disks across all nodes.
   - **Disks on selected nodes**: Uses matching disks only on the nodes you choose.
8. From **Disk Type**, select **SSD/NVMe** from the list.
9. Expand the **Advanced** section and configure the following:
   a. **Volume Mode**: Ensure that **Block** is selected.
   b. **Device Type**: Select one or more device types.
   c. **Disk Size**: Set the minimum and maximum disk size to include.
   d. **Maximum Disks Limit**: Specify the maximum number of PVs that can be created per node.



NOTE

If this field is empty, PVs are created for all matching disks.

10. Click **Create**.
11. Wait for the newly created PVs associated with the new LocalVolumeSet to become available.

Verification

- Verify that the local volume set is created:

```
$ oc get localvolumeset -n openshift-local-storage
NAME          AGE
localblock    16h
localvolume2  43m
```

- Verify the local storage class

```
oc get sc
NAME                        PROVISIONER                             RECLAIMPOLICY
VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSION   AGE
localblock                  kubernetes.io/no-provisioner            Delete
WaitForFirstConsumer   false                  15h
localvolume2                kubernetes.io/no-provisioner            Delete
WaitForFirstConsumer   false                  27m
ocs-storagecluster-ceph-rbd   openshift-storage.rbd.csi.ceph.com      Delete          Immediate
true                   15h
[...]
```

- Verify the PV by waiting for it to be available and it must be using the new storage class **localvolume2**:
  For example:

```
$ oc get pv | grep localvolume2
local-pv-14c0b1d                    130Gi      RWO          Delete          Available
localvolume2            <unset>                          8m55s
local-pv-41d0d077                   130Gi      RWO          Delete          Available
localvolume2            <unset>                          7m24s
local-pv-6c57a345                   130Gi      RWO          Delete          Available
localvolume2            <unset>                          5m4s
```

5.2. ATTACHING STORAGE FOR A NEW DEVICE SET

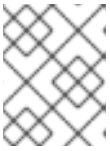
Prerequisites

- Make sure to label the required nodes in the new Local Volume Set with **cluster.ocs.openshift.io/openshift-storage=".**

```
oc label node <node-name > cluster.ocs.openshift.io/openshift-storage=" --overwrite
```

Procedure

1. In the OpenShift Web Console, navigate to **Storage → Storage cluster**.
2. Click **Actions** at the top right of the page and select **Attach Storage**.
3. From **LSO StorageClass**, select the newly created local storage class.
4. To enable encryption for the device set, select **Enable encryption on device set**



NOTE

Enabling or disabling OSD encryption using this option overrides the cluster-wide encryption setting for these new OSD storage devices.

5. Enter a name for the **Device class**.
   This is a label applied to OSDs to distinguish this set of devices from others (for example, by size, type, or node group). This option is available only when creating new devices of the same size on the same set of storage devices.

6. Select the **Volume type**.

7. Enter a name for the **Pool**.

8. Select the **Data protection policy**.
   It is recommended to choose **3-way Replication**.

9. Select the **Data compression** option to enable compression within replicas for improved storage efficiency.

10. Select the **Reclaim Policy**.

11. Select the **Volume Binding Mode**.

12. Enter a name for the **New User Storage Class**.

13. To generate an encryption key for each persistent volume created through this storage class, select **Enable Encryption on StorageClass**.

14. Click **Attach**.

Verification

- Verify that the PV and PVCs are in **Bound** state.
  For example:

```
$ oc get pv | grep localvolume2
local-pv-14c0b1d                           130Gi      RWO            Delete           Bound    openshift-
storage/localvolume2-0-data-0kp29f                localvolume2                <unset>
31m
local-pv-41d0d077                          130Gi      RWO            Delete           Bound    openshift-
storage/localvolume2-0-data-2vwk54                localvolume2                <unset>
30m
local-pv-6c57a345                          130Gi      RWO            Delete           Bound    openshift-
storage/localvolume2-0-data-1255ts                localvolume2                <unset>
28m
```

```
$ oc get pvc | grep localvolume2
localvolume2-0-data-0kp29f          Bound    local-pv-14c0b1d                  130Gi
RWO            localvolume2                <unset>                19m
localvolume2-0-data-1255ts          Bound    local-pv-6c57a345                 130Gi
RWO            localvolume2                <unset>                19m
localvolume2-0-data-2vwk54          Bound    local-pv-41d0d077                 130Gi
RWO            localvolume2                <unset>                19m
```

- Verify that the new OSDs are created successfully and all the OSDs are running.
  For example:

```
$ oc -n openshift-storage get pods -l app=rook-ceph-osd
NAME                           READY   STATUS    RESTARTS   AGE
rook-ceph-osd-0-7899b89478-bmg2l   2/2     Running   2          16h
rook-ceph-osd-1-6d7df8dbfc-z29jx   2/2     Running   2          16h
rook-ceph-osd-2-66b9dc8cd7-vd77d   2/2     Running   2          16h
rook-ceph-osd-3-79b59f44d7-cb9m7   2/2     Running   0          15m
rook-ceph-osd-4-766bcd8646-z4zvz   2/2     Running   0          15m
rook-ceph-osd-5-845554d78-qqcm4    2/2     Running   0          15m
```

- Verify that the storagecluster is in **Ready** state and the Ceph cluster health is **OK**.
  For example:

```
$ oc -n openshift-storage get storagecluster
NAME                 AGE    PHASE         EXTERNAL   CREATED AT             VERSION
ocs-storagecluster   2d2h   Ready                    2025-02-11T12:03:37Z   4.22.0

$ oc -n openshift-storage get cephcluster
NAME                             DATADIRHOSTPATH   MONCOUNT   AGE    PHASE   MESSAGE
HEALTH      EXTERNAL   FSID
ocs-storagecluster-cephcluster   /var/lib/rook     3          2d2h   Ready   Cluster created
successfully   HEALTH_OK            8ec4e898-5386-42c7-b01b-169fe8f08ba4
```

- In the Ceph cluster, check the Ceph OSD tree to see if the new device classes are spread correctly and OSDs are **up**.
  For example:

```
$ oc rsh -n openshift-storage $(oc get pods -o wide -n openshift-storage|grep tool|awk
'{print$1}') ceph osd tree
ID  CLASS        WEIGHT   TYPE NAME          STATUS  REWEIGHT  PRI-AFF
-1               0.67406  root default
-7               0.22469      host compute-0
 3  localvolume2  0.12700          osd.3          up   1.00000  1.00000
 0          ssd  0.09769          osd.0          up   1.00000  1.00000
-3               0.22469      host compute-1
 4  localvolume2  0.12700          osd.4          up   1.00000  1.00000
 2          ssd  0.09769          osd.2          up   1.00000  1.00000
-5               0.22469      host compute-2
 5  localvolume2  0.12700          osd.5          up   1.00000  1.00000
 1          ssd  0.09769          osd.1          up   1.00000  1.00000
```

- Verify that the user storageclass is created.
  For example:

```
$ oc get sc
NAME                          PROVISIONER                             RECLAIMPOLICY
VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSION   AGE
localblock                    kubernetes.io/no-provisioner            Delete
WaitForFirstConsumer   false                  16h
localvolume2                  kubernetes.io/no-provisioner            Delete
WaitForFirstConsumer   false                  63m
ocs-storagecluster-ceph-rbd   openshift-storage.rbd.csi.ceph.com      Delete          Immediate
```

```
true                    16h
ocs-storagecluster-ceph-rgw   openshift-storage.ceph.rook.io/bucket   Delete          Immediate
false                   16h
ocs-storagecluster-cephfs     openshift-storage.cephfs.csi.ceph.com   Delete          Immediate
true                    16h
openshift-storage.noobaa.io   openshift-storage.noobaa.io/obc         Delete          Immediate
false                   16h
ssd2                          openshift-storage.rbd.csi.ceph.com      Delete
WaitForFirstConsumer   true                    16m
thin-csi                      csi.vsphere.vmware.com                  Delete          WaitForFirstConsumer
true                    17h
thin-csi-odf                  csi.vsphere.vmware.com                  Delete
WaitForFirstConsumer   true                    16h
```

5.3. ATTACHING STORAGE FOR A NEW DEVICE SET BY USING DISKS OF SAME TYPE ON THE SAME NODES

Use this section when creating new devices of the same size on the same set of storage devices where you need to define a custom device class.

Pools write data only to OSDs that match the selected device class. By default, all OSDs use the **ssd** device class, so all pools can use all OSDs. If a custom device class is set, only new pools that explicitly use that device class will write to those OSDs. Existing pools will continue using only the OSDs that match their original device class.

Prerequisites

- Ensure that the worker nodes hosting the OSDs have a sufficient number of available disks.
- Make sure to label the required nodes in a new Local Volume Set with **cluster.ocs.openshift.io/openshift-storage=".**

  ```
  $ oc label node <node-name > cluster.ocs.openshift.io/openshift-storage=" --overwrite
  ```

- Ensure that an adequate number of Persistent Volumes (PVs) are available to match the number of disks that can be used for OSD provisioning.
  To verify the available local block PVs, run the following command:

  ```
  $ oc get pv | grep localblock
  ```

  Example output:

  ```
  $ oc get pv | grep localblock
  local-pv-1217fe9b                          100Gi      RWO            Delete           Available
  localblock              <unset>                          4m22s
  local-pv-18699823                          100Gi      RWO            Delete           Bound       openshift-
  storage/ocs-deviceset-localblock-0-data-122gpk        localblock              <unset>
  116m
  local-pv-7357d060                          100Gi      RWO            Delete           Available
  localblock              <unset>                          9m9s
  local-pv-c31f65ba                          100Gi      RWO            Delete           Available
  localblock              <unset>                          2m2s
  local-pv-d666f718                          100Gi      RWO            Delete           Bound       openshift-
  storage/ocs-deviceset-localblock-0-data-0cgff6        localblock              <unset>
  ```

```
151m
local-pv-fa54f87b 100Gi RWO Delete Bound openshift-
storage/ocs-deviceset-localblock-0-data-2d277g localblock <unset>
151m
```

In this example, there are three available disks, which correspond to three PVs in the Available state.

Procedure

1. In the OpenShift Web Console, navigate to **Storage → Storage cluster**.
2. Click **Actions** at the top right of the page and select **Attach Storage**.
3. From **LSO StorageClass**, select the newly created local storage class.
4. To enable encryption for the device set, select **Enable encryption on device set**



NOTE

Enabling or disabling OSD encryption using this option overrides the cluster-wide encryption setting for these new OSD storage devices.

5. Enter a name for the **Device class**.
   This is a label applied to OSDs to distinguish this set of devices from others (for example, by size, type, or node group). This option is available only when creating new devices of the same size on the same set of storage devices.
6. Select the **Volume type**.
7. Enter a name for the **Pool**.
8. Select the **Data protection policy**.
   It is recommended to choose **3-way Replication**.
9. Select the **Data compression** option to enable compression within replicas for improved storage efficiency.
10. Select the **Reclaim Policy**.
11. Select the **Volume Binding Mode**.
12. Enter a name for the **New User Storage Class**.
13. To generate an encryption key for each persistent volume created through this storage class, select **Enable encryption on StorageClass**.
14. Click **Attach**.

Verification

- Verify that the PV and PVCs are in **Bound** state.
  For example:

```
$ oc get pv | grep localblock-1
```

```
local-pv-1217fe9b                          100Gi      RWO            Delete           Bound    openshift-
storage/localblock-1-0-data-28ptwg                    localblock                  <unset>
84m
local-pv-7357d060                          100Gi      RWO            Delete           Bound    openshift-
storage/localblock-1-0-data-15td2h                    localblock                  <unset>
89m
local-pv-c31f65ba                          100Gi      RWO            Delete           Bound    openshift-
storage/localblock-1-0-data-08bjsj                    localblock                  <unset>
82m
```

```
$ oc get pvc | grep localblock-1
localblock-1-0-data-08bjsj              Bound    local-pv-c31f65ba                          100Gi      RWO
localblock                  <unset>                 79m
localblock-1-0-data-15td2h              Bound    local-pv-7357d060                          100Gi
RWO            localblock                  <unset>                 79m
localblock-1-0-data-28ptwg              Bound    local-pv-1217fe9b                          100Gi
RWO            localblock                  <unset>                 79m
```

- Verify that the new OSDs are created successfully and all the OSDs are running.
  For example:

```
$ oc get pods -n openshift-storage -l app=rook-ceph-osd
NAME                               READY   STATUS    RESTARTS   AGE
rook-ceph-osd-0-85f6969cff-l67mz   2/2     Running   2          173m
rook-ceph-osd-1-69c8f7c797-bvvpg   2/2     Running   2          172m
rook-ceph-osd-2-5cbc9cc8cb-rjmbv   2/2     Running   2          172m
rook-ceph-osd-3-76c84945b4-4mdv8   2/2     Running   0          84m
rook-ceph-osd-4-78c6b8bf87-gkh52   2/2     Running   0          84m
rook-ceph-osd-5-7876b5fb6f-m5j9q   2/2     Running   0          84m
```

- Verify that the storagecluster is in **Ready** state and the Ceph cluster health is **OK**.
  For example:

```
$ oc -n openshift-storage get storagecluster
NAME                 AGE     PHASE   EXTERNAL   CREATED AT             VERSION
ocs-storagecluster   4h34m   Ready              2026-03-04T12:17:21Z   4.22.0
```

```
$ oc -n openshift-storage get cephcluster
NAME                             DATADIRHOSTPATH   MONCOUNT   AGE    PHASE   MESSAGE
HEALTH      EXTERNAL   FSID
ocs-storagecluster-cephcluster   /var/lib/rook     3          4h34m   Ready   Cluster created
successfully   HEALTH_OK            0daaeaae-d8cd-45e6-9df8-f23fe3a70263
```

- In the Ceph cluster, check the Ceph OSD tree to see if the new device classes are spread correctly and OSDs are **up**.
  For example:

```
$ oc rsh -n openshift-storage $(oc get pods -o wide -n openshift-storage|grep tool|awk
'{print$1}') ceph osd tree
ID  CLASS        WEIGHT   TYPE NAME          STATUS  REWEIGHT  PRI-AFF
-1               0.58612  root default
-5               0.19537      host compute-0
 4  localblock-1  0.09769          osd.4          up   1.00000  1.00000
```

```
 1          ssd  0.09769         osd.1          up   1.00000  1.00000
-7                0.19537      host compute-1
 5  localblock-1  0.09769         osd.5          up   1.00000  1.00000
 2          ssd  0.09769         osd.2          up   1.00000  1.00000
-3                0.19537      host compute-2
 3  localblock-1  0.09769         osd.3          up   1.00000  1.00000
 0          ssd  0.09769         osd.0          up   1.00000  1.00000
```

- Verify that the user storageclass is created.
  For example:

```
$ oc get sc
NAME                          PROVISIONER                             RECLAIMPOLICY
VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSION   AGE
localblock                    kubernetes.io/no-provisioner            Delete
WaitForFirstConsumer   false                  5h30m
localblock-1-sc               openshift-storage.rbd.csi.ceph.com      Delete
WaitForFirstConsumer   true                   176m
ocs-storagecluster-ceph-rbd   openshift-storage.rbd.csi.ceph.com      Delete          Immediate
true                   4h25m
ocs-storagecluster-ceph-rgw   openshift-storage.ceph.rook.io/bucket   Delete          Immediate
false                  4h31m
ocs-storagecluster-cephfs     openshift-storage.cephfs.csi.ceph.com   Delete          Immediate
true                   4h25m
openshift-storage.noobaa.io   openshift-storage.noobaa.io/obc         Delete          Immediate
false                  4h18m
thin-csi                      csi.vsphere.vmware.com                  Delete          WaitForFirstConsumer
true                   6h8m
thin-csi-odf                  csi.vsphere.vmware.com                  Delete
WaitForFirstConsumer   true                   4h31m
```

CHAPTER 6. SCALING STORAGE OF VMWARE OPENSHIFT DATA FOUNDATION CLUSTER

6.1. SCALING UP STORAGE ON A DYNAMICALLY PROVISIONED VMWARE CLUSTER

To increase the storage capacity in a dynamically created storage cluster on a VMware user-provisioned infrastructure, you can add storage capacity and performance to your configured Red Hat OpenShift Data Foundation worker nodes.

Prerequisites

- Administrative privilege to the OpenShift Container Platform Console.
- A running OpenShift Data Foundation Storage Cluster.
- Make sure that the disk is of the same size and type as the disk used during initial deployment.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Storage → Storage cluster**.
    a. Click **Actions** at the top right of the page to extend the options menu.
    b. Select **Add Capacity** from the options menu.
    c. Select the **Storage Class**. Choose the storage class which you wish to use to provision new storage devices.
    d. Click **Add**.
3. To check the status, navigate to **Storage → Storage cluster** and verify that **Storage Cluster** in the Status card has a green tick.

Verification

- Verify the **Raw Capacity** card.
    a. In the OpenShift Web Console, click **Storage → Storage cluster**.
    b. In the **Block and File** tab, check the **Raw Capacity** card.
    Note that the capacity increases based on your selections.



NOTE

The raw capacity does not take replication into account and shows the full capacity.

- Verify that the new OSDs and their corresponding new Persistent Volume Claims (PVCs) are created.
    - To view the state of the newly created OSDs:

a. Click **Workloads → Pods** from the OpenShift Web Console.

b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

- To view the state of the PVCs:

a. Click **Storage → Persistent Volume Claims** from the OpenShift Web Console.

b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

- Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSD devices are encrypted.

a. Identify the nodes where the new OSD pods are running.

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
```

<OSD-pod-name>

Is the name of the OSD pod.
For example:

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
```

Example output:

```
NODE
compute-1
```

b. For each of the nodes identified in the previous step, do the following:

i. Create a debug pod and open a chroot environment for the selected hosts.

```
$ oc debug node/<node-name>
```

<node-name>

Is the name of the node.

```
$ chroot /host
```

ii. Check for the **crypt** keyword beside the **ocs-deviceset** names.

```
$ lsblk
```



IMPORTANT

Cluster reduction is supported only with the Red Hat Support Team's assistance.

6.2. SCALING UP A CLUSTER CREATED USING LOCAL STORAGE DEVICES

To scale up an OpenShift Data Foundation cluster which was created using local storage devices, you need to add a new disk to the storage node. The new disks size must be of the same size as the disks used during the deployment because OpenShift Data Foundation does not support heterogeneous disks/OSDs.

For deployments having three failure domains, you can scale up the storage by adding disks in the multiples of three, with the same number of disks coming from nodes in each of the failure domains. For example, if we scale by adding six disks, two disks are taken from nodes in each of the three failure domains. If the number of disks is not in multiples of three, it will only consume the disk to the maximum in the multiple of three while the remaining disks remain unused.

For deployments having less than three failure domains, there is a flexibility to add any number of disks. Make sure to verify that flexible scaling is enabled. For information, refer to the Knowledgebase article Verify if flexible scaling is enabled .



NOTE

Flexible scaling features get enabled at the time of deployment and cannot be enabled or disabled later on.

Prerequisites

- Administrative privilege to the OpenShift Container Platform Console.
- A running OpenShift Data Foundation Storage Cluster.
- Make sure that the disks to be used for scaling are attached to the storage node
- Make sure that **LocalVolumeDiscovery** and **LocalVolumeSet** objects are created.

To add capacity, you can either use a storage class that you provisioned during the deployment or any other storage class that matches the filter.

Procedure

1. In the OpenShift Web Console, click **Storage → Storage cluster**.
   a. Click **Actions** at the top right of the page to extend the options menu.
   b. Select **Add Capacity** from the options menu.

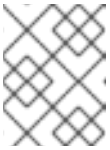
c. Select the **Storage Class** for which you added disks or the new storage class depending on your requirement. Available Capacity displayed is based on the local disks available in storage class.

d. Click **Add**.

2. To check the status, navigate to **Storage → Storage cluster** and verify that the **Storage Cluster** in the **Status** card has a green tick.

Verification

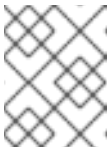
- Verify the **Raw Capacity** card.
  a. In the OpenShift Web Console, click **Storage → Storage cluster**.
  b. In the **Block and File** tab, check the **Raw Capacity** card.
  Note that the capacity increases based on your selections.



NOTE

The raw capacity does not take replication into account and shows the full capacity.

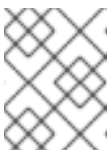
- Verify that the new OSDs and their corresponding new Persistent Volume Claims (PVCs) are created.
  - To view the state of the newly created OSDs:
    a. Click **Workloads → Pods** from the OpenShift Web Console.
    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

  - To view the state of the PVCs:
    a. Click **Storage → Persistent Volume Claims** from the OpenShift Web Console.
    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

- Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSD devices are encrypted.
  a. Identify the nodes where the new OSD pods are running.

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
```

<OSD-pod-name>

Is the name of the OSD pod.
For example:

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
```

Example output:

```
NODE
compute-1
```

b. For each of the nodes identified in the previous step, do the following:

i. Create a debug pod and open a chroot environment for the selected host(s).

```
$ oc debug node/<node-name>
```

<node-name>

Is the name of the node.

```
$ chroot /host
```

ii. Check for the **crypt** keyword beside the **ocs-deviceset** names.

```
$ lsblk
```



IMPORTANT

Cluster reduction is supported only with the Red Hat Support Team's assistance.

6.2.1. Scaling up storage capacity on a cluster by resizing existing OSDs

To increase the storage capacity on a cluster, you can add storage capacity by resizing existing OSDs.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Update the **dataPVCTemplate** size for the **storageDeviceSets** in the StorageCluster CR with the new desired size. For example:

```
apiVersion: ocs.openshift.io/v1
kind: StorageCluster
metadata:
  namespace: openshift-storage
```

```
  name: example-storagecluster
spec:
  manageNodes: false
  monPVCTemplate:
    spec:
      storageClassName:
      accessModes:
      - ReadWriteOnce
      resources:
        requests:
          storage: 10Gi
  storageDeviceSets:
  - name: example-deviceset
    count: 3
    resources: {}
    placement: {}
    dataPVCTemplate:
      spec:
        storageClassName:
        accessModes:
        - ReadWriteOnce
        volumeMode: Block
        resources:
          requests:
            storage: 1Ti
    portable: true
```

In this example YAML, the **storage** parameter under **storageDeviceSets** reflects the size change of **1Ti**.

2. Wait for the OSDs to restart.

3. Confirm that the resize took effect:

```
$ oc get pvc -l ceph.rook.io/DeviceSet
```

Example output:

```
% oc get pvc -l ceph.rook.io/DeviceSet
NAME                STATUS   VOLUME                                     CAPACITY   ACCESS MODES
STORAGECLASS                  AGE
rook-ceph-osd-a4681   Bound    pvc-c3868d38-6bd9-4c8d-91e1-a4681e4cb8c1   1Ti
RWO            gp2-csi   176m
```

Verify the resize is complete in the **CAPACITY** column.

4. If the resize did not take effect, restart the OSD pods again. It may take multiple restarts for the resize to complete.

6.3. ENABLING AUTOMATIC CAPACITY SCALING

You can enable automatic capacity scaling on clusters deployed using dynamic storage devices. When automatic capacity scaling is enabled, additional raw capacity equivalent to the configured deployment size is automatically added to the cluster when used capacity reaches 70%. This ensures your deployment scales seamlessly to meet demand.

This option is disabled in lean profile mode, LSO deployment, and external mode deployment.



IMPORTANT

This may incur additional costs for the underlying storage.

Prerequisites

- Administrative privilege to the OpenShift Container Platform Console.
- A running OpenShift Data Foundation Storage Cluster.



IMPORTANT

IBM cloud **10iops-tier** workloads are not supported.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Storage → Storage cluster**.
3. Click **Actions** at the top right of the page to extend the options menu.
4. Select **Automatic capacity scaling** from the options menu.
5. In the **Automatic capacity scaling** page, select the **Enable automatic capacity scaling for your cluster** checkbox.
6. Set the cluster expansion limit from the dropdown. This is the maximum the cluster can expand in the cloud. Automatic scaling is suspended if this limit is exceeded.
7. Click **Save changes**.

6.4. SCALING OUT STORAGE CAPACITY ON A VMWARE CLUSTER

6.4.1. Adding a node to an installer-provisioned infrastructure

Use this procedure to add a new node to an installer-provisioned infrastructure cluster.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

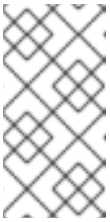
1. Navigate to **Compute → Machine Sets**.
2. On the machine set where you want to add nodes, select **Edit Machine Count**
   a. Add the amount of nodes, and click **Save**.

b. Click **Compute → Nodes** and confirm if the new node is in **Ready** state.

3. Apply the OpenShift Data Foundation label to the new node.

a. For the new node, click **Action menu (⋮) → Edit Labels**.

b. Add **cluster.ocs.openshift.io/openshift-storage**, and click **Save**.



NOTE

It is recommended to add 3 nodes, one each in different zones. You must add 3 nodes and perform this procedure for all of them. In case of bare metal installer-provisioned infrastructure deployment, you must expand the cluster first. For instructions, see Expanding the cluster.

Verification

1. Execute the following command in the terminal and verify that the new node is present in the output:

```
$ oc get nodes --show-labels | grep cluster.ocs.openshift.io/openshift-storage= |cut -d' ' -f1
```

2. On the OpenShift web console, click **Workloads → Pods**, confirm that at least the following pods on the new node are in **Running** state:

- **openshift-storage.cephfs.csi.ceph.com-***
- **openshift-storage.rbd.csi.ceph.com-***

6.4.2. Adding a node to an user-provisioned infrastructure

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Depending on the type of infrastructure, perform the following steps:

a. Get a new machine with the required infrastructure. See Platform requirements.

b. Create a new OpenShift Container Platform worker node using the new machine.

2. Check for certificate signing requests (CSRs) that are in **Pending** state.

```
$ oc get csr
```

3. Approve all the required CSRs for the new node.

```
$ oc adm certificate approve <Certificate_Name>
```

<Certificate_Name>

Is the name of the CSR.

4. Click **Compute → Nodes**, confirm if the new node is in **Ready** state.
5. Apply the OpenShift Data Foundation label to the new node using any one of the following:

   From User interface

   a. For the new node, click **Action Menu (⋮) → Edit Labels**.

   b. Add **cluster.ocs.openshift.io/openshift-storage**, and click **Save**.

   From Command line interface

   - Apply the OpenShift Data Foundation label to the new node.

     ```
     $ oc label node <new_node_name> cluster.ocs.openshift.io/openshift-storage=""
     ```

     <new_node_name>

     Is the name of the new node.

Verification

1. Execute the following command in the terminal and verify that the new node is present in the output:

   ```
   $ oc get nodes --show-labels | grep cluster.ocs.openshift.io/openshift-storage= |cut -d' ' -f1
   ```

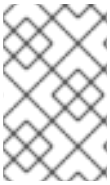
2. On the OpenShift web console, click **Workloads → Pods**, confirm that at least the following pods on the new node are in **Running** state:

   - **openshift-storage.cephfs.csi.ceph.com-***
   - **openshift-storage.rbd.csi.ceph.com-***

6.4.3. Adding a node using a local storage device

You can add nodes to increase the storage capacity when existing worker nodes are already running at their maximum supported OSDs or when there are not enough resources to add new OSDs on the existing nodes.

Add nodes in the multiple of 3, each of them in different failure domains. Though it is recommended to add nodes in multiples of 3 nodes, you have the flexibility to add one node at a time in flexible scaling deployment. See Knowledgebase article Verify if flexible scaling is enabled



NOTE

OpenShift Data Foundation does not support heterogeneous disk size and types. The new nodes to be added should have the disk of the same type and size which was used during initial OpenShift Data Foundation deployment.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.

- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Depending on the type of infrastructure, perform the following steps:
   a. Get a new machine with the required infrastructure. See Platform requirements.
   b. Create a new OpenShift Container Platform worker node using the new machine.
2. Check for certificate signing requests (CSRs) that are in **Pending** state.

```
$ oc get csr
```

3. Approve all the required CSRs for the new node.

```
$ oc adm certificate approve <Certificate_Name>
```

<Certificate_Name>
Is the name of the CSR.

4. Click **Compute → Nodes**, confirm if the new node is in **Ready** state.
5. Apply the OpenShift Data Foundation label to the new node using any one of the following:

From User interface

   a. For the new node, click **Action Menu (⋮) → Edit Labels**.
   b. Add **cluster.ocs.openshift.io/openshift-storage**, and click **Save**.

From Command line interface

- Apply the OpenShift Data Foundation label to the new node.

```
$ oc label node <new_node_name> cluster.ocs.openshift.io/openshift-storage=""
```

<new_node_name>
Is the name of the new node.

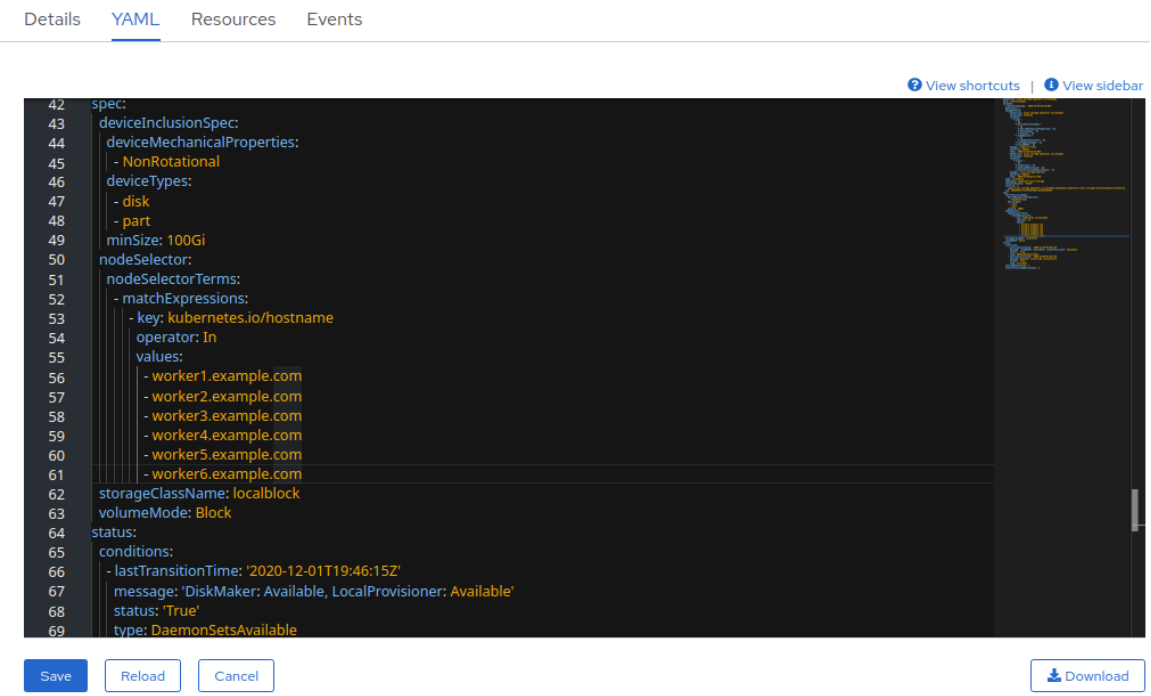
6. Click **Ecosystem → Installed Operators** from the OpenShift Web Console.
   From the **Project** drop-down list, make sure to select the project where the Local Storage Operator is installed.
7. Click **Local Storage**.
8. Click the **Local Volume Discovery** tab.
   a. Beside the **LocalVolumeDiscovery**, click Action menu **(⋮) → Edit Local Volume Discovery**.
   b. In the YAML, add the hostname of the new node in the **values** field under the node selector.

c. Click **Save**.

9. Click the **Local Volume Sets** tab.

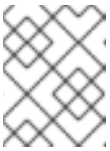
a. Beside the **LocalVolumeSet**, click Action menu (⋮) → **Edit Local Volume Set**

b. In the YAML, add the hostname of the new node in the **values** field under the **node selector**.



YAML showing the addition of new hostnames

c. Click **Save**.



NOTE

It is recommended to add 3 nodes, one each in different zones. You must add 3 nodes and perform this procedure for all of them.

Verification

1. Execute the following command in the terminal and verify that the new node is present in the output:

```
$ oc get nodes --show-labels | grep cluster.ocs.openshift.io/openshift-storage= |cut -d' ' -f1
```

2. On the OpenShift web console, click **Workloads → Pods**, confirm that at least the following pods on the new node are in **Running** state:

- **openshift-storage.cephfs.csi.ceph.com-***
- **openshift-storage.rbd.csi.ceph.com-***

6.4.4. Scaling up storage capacity

To scale up storage capacity:

- For dynamic storage devices, see Scaling up storage capacity on a cluster .
- For local storage devices, see Scaling up a cluster created using local storage devices

6.5. SCALING NON-RESILIENT POOLS

After enabling the single replica feature, you can increase the number of non-resilient pools by updating the pool count in the **StorageCluster** custom resource. Each additional pool is created in a separate failure domain and requires one new storage device per failure domain.



WARNING

Non-resilient pools have no data replication. Increasing the pool count does not provide data redundancy. Ensure that your applications manage their own replication before scaling.

Prerequisites

- The single replica feature is already enabled. For more information, see Storage class with single replica.
- For LSO-based deployments, ensure that the required number of unused local storage devices are correctly labeled on the nodes before scaling.

Procedure

1. Run the following command to increase the non-resilient pool count:

```
$ oc patch storagecluster ocs-storagecluster \
  -n openshift-storage \
  --type=json \
  --patch='[
    {
      "op": "replace",
      "path": "/spec/managedResources/cephNonResilientPools/count",
      "value": 2
    }
  ]'
```

   Replace **2** with the desired number of non-resilient pools. The value must not exceed the number of available failure domains.



NOTE

For LSO-based deployments, the operator creates one OSD per local device per failure domain. Ensure that the unused local devices are available on the nodes before running this command.

2. Verify that the **StorageCluster** is in **Ready** state:

```
$ oc get storagecluster -n openshift-storage
```

Example output:

```
NAME                 AGE   PHASE   EXTERNAL   CREATED AT             VERSION
ocs-storagecluster   10m   Ready              2024-02-05T13:56:15Z   4.22.0
```

3. Verify that the new **CephBlockPools** are created for each failure domain and are in **Ready** state:

```
$ oc get cephblockpools -n openshift-storage
```

Example output:

```
NAME                                      PHASE
ocs-storagecluster-cephblockpool          Ready
ocs-storagecluster-cephblockpool-us-east-1a   Ready
ocs-storagecluster-cephblockpool-us-east-1b   Ready
ocs-storagecluster-cephblockpool-us-east-1c   Ready
```

4. Verify that the new OSD pods are in **Running** state:

```
$ oc get pods -n openshift-storage | grep osd
```

CHAPTER 7. SCALING STORAGE OF MICROSOFT AZURE OPENSHIFT DATA FOUNDATION CLUSTER

To scale the storage capacity of your configured Red Hat OpenShift Data Foundation worker nodes on Microsoft Azure cluster, you can increase the capacity by adding three disks at a time. Three disks are needed since OpenShift Data Foundation uses a replica count of 3 to maintain the high availability. So the amount of storage consumed is three times the usable space.



NOTE

Usable space might vary when encryption is enabled or replica 2 pools are being used.

7.1. SCALING UP STORAGE CAPACITY ON A CLUSTER

To increase the storage capacity in a dynamically created storage cluster on an user-provisioned infrastructure, you can add storage capacity and performance to your configured Red Hat OpenShift Data Foundation worker nodes.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.
- The disk should be of the same size and type as used during initial deployment.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Storage → Storage cluster**.
    a. Click **Actions** at the top right of the page to extend the options menu.
    b. Select **Add Capacity** from the options menu.
    c. Select the **Storage Class**. Choose the storage class which you wish to use to provision new storage devices.
    d. Click **Add**.
3. To check the status, navigate to **Storage → Storage cluster** and verify that the **Storage Cluster** in the Status card has a green tick.

Verification

- Verify the **Raw Capacity** card.
    a. In the OpenShift Web Console, click **Storage → Storage cluster**.
    b. In the **Block and File** tab, check the **Raw Capacity** card.
    Note that the capacity increases based on your selections.



NOTE

The raw capacity does not take replication into account and shows the full capacity.

- Verify that the new object storage devices (OSDs) and their corresponding new Persistent Volume Claims (PVCs) are created.
  - To view the state of the newly created OSDs:
    a. Click **Workloads → Pods** from the OpenShift Web Console.
    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

  - To view the state of the PVCs:
    a. Click **Storage → Persistent Volume Claims** from the OpenShift Web Console.
    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

- Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSD devices are encrypted.
  a. Identify the nodes where the new OSD pods are running.

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
```

<OSD-pod-name>

Is the name of the OSD pod.
For example:

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
```

Example output:

```
NODE
compute-1
```

  b. For each of the nodes identified in the previous step, do the following:
    i. Create a debug pod and open a chroot environment for the selected hosts.

```
$ oc debug node/<node-name>
```

<node-name>

Is the name of the node.

```
$ chroot /host
```

ii. Check for the **crypt** keyword beside the **ocs-deviceset** names.

```
$ lsblk
```



IMPORTANT

Cluster reduction is supported only with the Red Hat Support Team's assistance.

7.1.1. Scaling up storage capacity on a cluster by resizing existing OSDs

To increase the storage capacity on a cluster, you can add storage capacity by resizing existing OSDs.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Update the **dataPVCTemplate** size for the **storageDeviceSets** in the StorageCluster CR with the new desired size. For example:

```
apiVersion: ocs.openshift.io/v1
kind: StorageCluster
metadata:
  namespace: openshift-storage
  name: example-storagecluster
spec:
  manageNodes: false
  monPVCTemplate:
    spec:
      storageClassName:
      accessModes:
      - ReadWriteOnce
      resources:
        requests:
          storage: 10Gi
  storageDeviceSets:
  - name: example-deviceset
    count: 3
    resources: {}
    placement: {}
    dataPVCTemplate:
      spec:
```

```
      storageClassName:
      accessModes:
      - ReadWriteOnce
      volumeMode: Block
      resources:
        requests:
          storage: 1Ti
    portable: true
```

In this example YAML, the **storage** parameter under **storageDeviceSets** reflects the size change of **1Ti**.

2. Wait for the OSDs to restart.

3. Confirm that the resize took effect:

```
$ oc get pvc -l ceph.rook.io/DeviceSet
```

Example output:

```
% oc get pvc -l ceph.rook.io/DeviceSet
NAME                STATUS   VOLUME                                     CAPACITY   ACCESS MODES
STORAGECLASS                  AGE
rook-ceph-osd-a4681   Bound    pvc-c3868d38-6bd9-4c8d-91e1-a4681e4cb8c1   1Ti
RWO            gp2-csi   176m
```

Verify the resize is complete in the **CAPACITY** column.

4. If the resize did not take effect, restart the OSD pods again. It may take multiple restarts for the resize to complete.

7.2. ENABLING AUTOMATIC CAPACITY SCALING

You can enable automatic capacity scaling on clusters deployed using dynamic storage devices. When automatic capacity scaling is enabled, additional raw capacity equivalent to the configured deployment size is automatically added to the cluster when used capacity reaches 70%. This ensures your deployment scales seamlessly to meet demand.

This option is disabled in lean profile mode, LSO deployment, and external mode deployment.

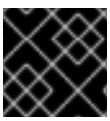


IMPORTANT

This may incur additional costs for the underlying storage.

Prerequisites

- Administrative privilege to the OpenShift Container Platform Console.
- A running OpenShift Data Foundation Storage Cluster.



IMPORTANT

IBM cloud **10iops-tier** workloads are not supported.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Storage → Storage cluster**.
3. Click **Actions** at the top right of the page to extend the options menu.
4. Select **Automatic capacity scaling** from the options menu.
5. In the **Automatic capacity scaling** page, select the **Enable automatic capacity scaling for your cluster** checkbox.
6. Set the cluster expansion limit from the dropdown. This is the maximum the cluster can expand in the cloud. Automatic scaling is suspended if this limit is exceeded.
7. Click **Save changes**.

7.3. SCALING OUT STORAGE CAPACITY ON A MICROSOFT AZURE CLUSTER

OpenShift Data Foundation is highly scalable. It can be scaled out by adding new nodes with required storage and enough hardware resources in terms of CPU and RAM. Practically there is no limit on the number of nodes which can be added but from the support perspective 2000 nodes is the limit for OpenShift Data Foundation.

Scaling out storage capacity can be broken down into two steps

- Adding new node
- Scaling up the storage capacity



NOTE

OpenShift Data Foundation does not support heterogeneous OSD/Disk sizes.

7.3.1. Adding a node to an installer-provisioned infrastructure

Use this procedure to add a new node to an installer-provisioned infrastructure cluster.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Navigate to **Compute → Machine Sets**.
2. On the machine set where you want to add nodes, select **Edit Machine Count**
   a. Add the amount of nodes, and click **Save**.
   b. Click **Compute → Nodes** and confirm if the new node is in **Ready** state.

3. Apply the OpenShift Data Foundation label to the new node.
   a. For the new node, click **Action menu (⋮) → Edit Labels**.
   b. Add **cluster.ocs.openshift.io/openshift-storage**, and click **Save**.



NOTE

It is recommended to add 3 nodes, one each in different zones. You must add 3 nodes and perform this procedure for all of them. In case of bare metal installer-provisioned infrastructure deployment, you must expand the cluster first. For instructions, see Expanding the cluster.

Verification

1. Execute the following command in the terminal and verify that the new node is present in the output:

```
$ oc get nodes --show-labels | grep cluster.ocs.openshift.io/openshift-storage= |cut -d' ' -f1
```

2. On the OpenShift web console, click **Workloads → Pods**, confirm that at least the following pods on the new node are in **Running** state:
   - **openshift-storage.cephfs.csi.ceph.com-***
   - **openshift-storage.rbd.csi.ceph.com-***

7.3.2. Scaling up storage capacity

To scale up storage capacity, see Scaling up storage capacity on a cluster .

7.4. SCALING NON-RESILIENT POOLS

After enabling the single replica feature, you can increase the number of non-resilient pools by updating the pool count in the **StorageCluster** custom resource. Each additional pool is created in a separate failure domain and requires one new storage device per failure domain.



WARNING

Non-resilient pools have no data replication. Increasing the pool count does not provide data redundancy. Ensure that your applications manage their own replication before scaling.

Prerequisites

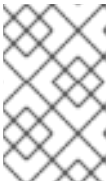
- The single replica feature is already enabled. For more information, see Storage class with single replica.
- For LSO-based deployments, ensure that the required number of unused local storage devices are correctly labeled on the nodes before scaling.

Procedure

1. Run the following command to increase the non-resilient pool count:

```
$ oc patch storagecluster ocs-storagecluster \
  -n openshift-storage \
  --type=json \
  --patch='[
    {
      "op": "replace",
      "path": "/spec/managedResources/cephNonResilientPools/count",
      "value": 2
    }
  ]'
```

Replace **2** with the desired number of non-resilient pools. The value must not exceed the number of available failure domains.



NOTE

For LSO-based deployments, the operator creates one OSD per local device per failure domain. Ensure that the unused local devices are available on the nodes before running this command.

2. Verify that the **StorageCluster** is in **Ready** state:

```
$ oc get storagecluster -n openshift-storage
```

Example output:

```
NAME                 AGE   PHASE   EXTERNAL   CREATED AT             VERSION
ocs-storagecluster   10m   Ready              2024-02-05T13:56:15Z   4.22.0
```

3. Verify that the new **CephBlockPools** are created for each failure domain and are in **Ready** state:

```
$ oc get cephblockpools -n openshift-storage
```

Example output:

```
NAME                                      PHASE
ocs-storagecluster-cephblockpool          Ready
ocs-storagecluster-cephblockpool-us-east-1a   Ready
ocs-storagecluster-cephblockpool-us-east-1b   Ready
ocs-storagecluster-cephblockpool-us-east-1c   Ready
```

4. Verify that the new OSD pods are in **Running** state:

```
$ oc get pods -n openshift-storage | grep osd
```

CHAPTER 8. SCALING STORAGE CAPACITY OF GCP OPENSHIFT DATA FOUNDATION CLUSTER

To scale the storage capacity of your configured Red Hat OpenShift Data Foundation worker nodes on GCP cluster, you can increase the capacity by adding three disks at a time. Three disks are needed since OpenShift Data Foundation uses a replica count of 3 to maintain the high availability. So the amount of storage consumed is three times the usable space.



NOTE

Usable space might vary when encryption is enabled or replica 2 pools are being used.

8.1. SCALING UP STORAGE CAPACITY ON A CLUSTER

To increase the storage capacity in a dynamically created storage cluster on an user-provisioned infrastructure, you can add storage capacity and performance to your configured Red Hat OpenShift Data Foundation worker nodes.

Prerequisites

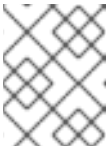
- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.
- The disk should be of the same size and type as used during initial deployment.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Storage → Storage cluster**.
    a. Click **Actions** at the top right of the page to extend the options menu.
    b. Select **Add Capacity** from the options menu.
    c. Select the **Storage Class**. Choose the storage class which you wish to use to provision new storage devices.
    d. Click **Add**.
3. To check the status, navigate to **Storage → Storage cluster** and verify that the **Storage Cluster** in the Status card has a green tick.

Verification

- Verify the **Raw Capacity** card.
    a. In the OpenShift Web Console, click **Storage → Storage cluster**.
    b. In the **Block and File** tab, check the **Raw Capacity** card.
    Note that the capacity increases based on your selections.



NOTE

The raw capacity does not take replication into account and shows the full capacity.

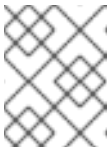
- Verify that the new object storage devices (OSDs) and their corresponding new Persistent Volume Claims (PVCs) are created.
  - To view the state of the newly created OSDs:
    a. Click **Workloads → Pods** from the OpenShift Web Console.
    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

  - To view the state of the PVCs:
    a. Click **Storage → Persistent Volume Claims** from the OpenShift Web Console.
    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

- Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSD devices are encrypted.
  a. Identify the nodes where the new OSD pods are running.

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
```

<OSD-pod-name>

Is the name of the OSD pod.
For example:

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
```

Example output:

```
NODE
compute-1
```

  b. For each of the nodes identified in the previous step, do the following:
    i. Create a debug pod and open a chroot environment for the selected hosts.

```
$ oc debug node/<node-name>
```

<node-name>

Is the name of the node.

```
$ chroot /host
```

ii. Check for the **crypt** keyword beside the **ocs-deviceset** names.

```
$ lsblk
```



IMPORTANT

Cluster reduction is supported only with the Red Hat Support Team's assistance.

8.1.1. Scaling up storage capacity on a cluster by resizing existing OSDs

To increase the storage capacity on a cluster, you can add storage capacity by resizing existing OSDs.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Update the **dataPVCTemplate** size for the **storageDeviceSets** in the StorageCluster CR with the new desired size. For example:

```
apiVersion: ocs.openshift.io/v1
kind: StorageCluster
metadata:
  namespace: openshift-storage
  name: example-storagecluster
spec:
  manageNodes: false
  monPVCTemplate:
    spec:
      storageClassName:
      accessModes:
      - ReadWriteOnce
      resources:
        requests:
          storage: 10Gi
  storageDeviceSets:
  - name: example-deviceset
    count: 3
    resources: {}
    placement: {}
    dataPVCTemplate:
      spec:
```

```
      storageClassName:
      accessModes:
      - ReadWriteOnce
      volumeMode: Block
      resources:
        requests:
          storage: 1Ti
    portable: true
```

In this example YAML, the **storage** parameter under **storageDeviceSets** reflects the size change of **1Ti**.

2. Wait for the OSDs to restart.
3. Confirm that the resize took effect:

```
$ oc get pvc -l ceph.rook.io/DeviceSet
```

Example output:

```
% oc get pvc -l ceph.rook.io/DeviceSet
NAME                STATUS   VOLUME                                     CAPACITY   ACCESS MODES
STORAGECLASS                  AGE
rook-ceph-osd-a4681   Bound    pvc-c3868d38-6bd9-4c8d-91e1-a4681e4cb8c1   1Ti
RWO            gp2-csi   176m
```

Verify the resize is complete in the **CAPACITY** column.

4. If the resize did not take effect, restart the OSD pods again. It may take multiple restarts for the resize to complete.

8.2. ENABLING AUTOMATIC CAPACITY SCALING

You can enable automatic capacity scaling on clusters deployed using dynamic storage devices. When automatic capacity scaling is enabled, additional raw capacity equivalent to the configured deployment size is automatically added to the cluster when used capacity reaches 70%. This ensures your deployment scales seamlessly to meet demand.

This option is disabled in lean profile mode, LSO deployment, and external mode deployment.

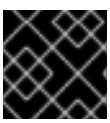


IMPORTANT

This may incur additional costs for the underlying storage.

Prerequisites

- Administrative privilege to the OpenShift Container Platform Console.
- A running OpenShift Data Foundation Storage Cluster.



IMPORTANT

IBM cloud **10iops-tier** workloads are not supported.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Storage → Storage cluster**.
3. Click **Actions** at the top right of the page to extend the options menu.
4. Select **Automatic capacity scaling** from the options menu.
5. In the **Automatic capacity scaling** page, select the **Enable automatic capacity scaling for your cluster** checkbox.
6. Set the cluster expansion limit from the dropdown. This is the maximum the cluster can expand in the cloud. Automatic scaling is suspended if this limit is exceeded.
7. Click **Save changes**.

8.3. SCALING OUT STORAGE CAPACITY ON A GCP CLUSTER

OpenShift Data Foundation is highly scalable. It can be scaled out by adding new nodes with required storage and enough hardware resources in terms of CPU and RAM. Practically there is no limit on the number of nodes which can be added but from the support perspective 2000 nodes is the limit for OpenShift Data Foundation.

Scaling out storage capacity can be broken down into two steps

- Adding new node
- Scaling up the storage capacity



NOTE

OpenShift Data Foundation does not support heterogeneous OSD/Disk sizes.

8.3.1. Adding a node

You can add nodes to increase the storage capacity when existing worker nodes are already running at their maximum supported OSDs or there are not enough resources to add new OSDs on the existing nodes. It is always recommended to add nodes in the multiple of three, each of them in different failure domains.

While it is recommended to add nodes in the multiple of three, you still have the flexibility to add one node at a time in the flexible scaling deployment. Refer to the Knowledgebase article Verify if flexible scaling is enabled.



NOTE

OpenShift Data Foundation does not support heterogeneous disk size and types. The new nodes to be added should have the disk of the same type and size which was used during OpenShift Data Foundation deployment.

8.3.1.1. Adding a node to an installer-provisioned infrastructure

Use this procedure to add a new node to an installer-provisioned infrastructure cluster.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Navigate to **Compute → Machine Sets**.
2. On the machine set where you want to add nodes, select **Edit Machine Count**
   a. Add the amount of nodes, and click **Save**.
   b. Click **Compute → Nodes** and confirm if the new node is in **Ready** state.
3. Apply the OpenShift Data Foundation label to the new node.
   a. For the new node, click **Action menu (⋮) → Edit Labels**.
   b. Add **cluster.ocs.openshift.io/openshift-storage**, and click **Save**.



NOTE

It is recommended to add 3 nodes, one each in different zones. You must add 3 nodes and perform this procedure for all of them. In case of bare metal installer-provisioned infrastructure deployment, you must expand the cluster first. For instructions, see Expanding the cluster.

Verification

1. Execute the following command in the terminal and verify that the new node is present in the output:

```
$ oc get nodes --show-labels | grep cluster.ocs.openshift.io/openshift-storage= |cut -d' ' -f1
```

2. On the OpenShift web console, click **Workloads → Pods**, confirm that at least the following pods on the new node are in **Running** state:
   - **openshift-storage.cephfs.csi.ceph.com-***
   - **openshift-storage.rbd.csi.ceph.com-***

8.3.2. Scaling up storage capacity

To scale up storage capacity, see Scaling up storage capacity on a cluster .

8.4. SCALING NON-RESILIENT POOLS

After enabling the single replica feature, you can increase the number of non-resilient pools by updating the pool count in the **StorageCluster** custom resource. Each additional pool is created in a separate failure domain and requires one new storage device per failure domain.



WARNING

Non-resilient pools have no data replication. Increasing the pool count does not provide data redundancy. Ensure that your applications manage their own replication before scaling.

Prerequisites

- The single replica feature is already enabled. For more information, see Storage class with single replica.
- For LSO-based deployments, ensure that the required number of unused local storage devices are correctly labeled on the nodes before scaling.

Procedure

1. Run the following command to increase the non-resilient pool count:

```
$ oc patch storagecluster ocs-storagecluster \
  -n openshift-storage \
  --type=json \
  --patch='[
    {
      "op": "replace",
      "path": "/spec/managedResources/cephNonResilientPools/count",
      "value": 2
    }
  ]'
```

Replace **2** with the desired number of non-resilient pools. The value must not exceed the number of available failure domains.



NOTE

For LSO-based deployments, the operator creates one OSD per local device per failure domain. Ensure that the unused local devices are available on the nodes before running this command.

2. Verify that the **StorageCluster** is in **Ready** state:

```
$ oc get storagecluster -n openshift-storage
```

Example output:

```
NAME                 AGE   PHASE   EXTERNAL   CREATED AT             VERSION
ocs-storagecluster   10m   Ready              2024-02-05T13:56:15Z   4.22.0
```

3. Verify that the new **CephBlockPools** are created for each failure domain and are in **Ready** state:

```
$ oc get cephblockpools -n openshift-storage
```

Example output:

```
NAME                                      PHASE
ocs-storagecluster-cephblockpool          Ready
ocs-storagecluster-cephblockpool-us-east-1a   Ready
ocs-storagecluster-cephblockpool-us-east-1b   Ready
ocs-storagecluster-cephblockpool-us-east-1c   Ready
```

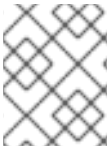
4. Verify that the new OSD pods are in **Running** state:

```
$ oc get pods -n openshift-storage | grep osd
```

CHAPTER 9. SCALING STORAGE OF IBM Z OR IBM LINUXONE OPENSHIFT DATA FOUNDATION CLUSTER

9.1. SCALING UP STORAGE BY ADDING CAPACITY TO YOUR OPENSHIFT DATA FOUNDATION NODES ON IBM Z OR IBM LINUXONE INFRASTRUCTURE

You can add storage capacity and performance to your configured Red Hat OpenShift Data Foundation worker nodes.



NOTE

Flexible scaling features get enabled at the time of deployment and can not be enabled or disabled later on.

Prerequisites

- A running OpenShift Data Foundation Platform.
- Administrative privileges on the OpenShift Web Console.
- To scale using a storage class other than the one provisioned during deployment, first define an additional storage class. See Creating storage classes and pools for details.

Procedure

1. Add additional hardware resources with zFCP disks.
   a. List all the disks.

```
$ lszdev
```

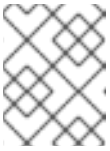
   Example output:

```
TYPE        ID                                              ON   PERS  NAMES
zfcp-host   0.0.8204                                        yes  yes
zfcp-lun    0.0.8204:0x102107630b1b5060:0x4001402900000000  yes  no    sda sg0
zfcp-lun    0.0.8204:0x500407630c0b50a4:0x3002b03000000000  yes  yes   sdb sg1
qeth        0.0.bdd0:0.0.bdd1:0.0.bdd2                      yes  no    encbdd0
generic-ccw 0.0.0009                                        yes  no
```

   A SCSI disk is represented as a **zfcp-lun** with the structure **<device-id>:<wwpn>:<lun-id>** in the ID section. The first disk is used for the operating system. The device id for the new disk can be the same.

   b. Append a new SCSI disk.

```
$ chzdev -e 0.0.8204:0x400506630b1b50a4:0x3001301a00000000
```



NOTE

The device ID for the new disk must be the same as the disk to be replaced. The new disk is identified with its WWPN and LUN ID.

c. List all the FCP devices to verify the new disk is configured.

```
$ lszdev zfcp-lun
TYPE       ID                                                  ON   PERS  NAMES
zfcp-lun   0.0.8204:0x102107630b1b5060:0x4001402900000000 yes  no    sda sg0
zfcp-lun   0.0.8204:0x500507630b1b50a4:0x4001302a00000000  yes  yes   sdb sg1
zfcp-lun   0.0.8204:0x400506630b1b50a4:0x3001301a00000000  yes  yes   sdc sg2
```

2. Navigate to the OpenShift Web Console.

3. Click **Storage → Storage cluster**.

a. Click **Actions** at the top right of the page to extend the options menu.

b. Select **Add Capacity** from the options menu.
The **Raw Capacity** field shows the size set during storage class creation. The total amount of storage consumed is three times this amount, because OpenShift Data Foundation uses a replica count of 3.

c. Click **Add**.

1. To check the status, navigate to **Storage → Storage cluster** and verify that **Storage Cluster** in the **Status** card has a green tick.

Verification

- Verify the **Raw Capacity** card.

  a. In the OpenShift Web Console, click **Storage → Storage cluster**.

  b. In the **Block and File** tab, check the **Raw Capacity** card.
  Note that the capacity increases based on your selections.



NOTE

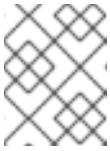
The raw capacity does not take replication into account and shows the full capacity.

- Verify that the new OSDs and their corresponding new Persistent Volume Claims (PVCs) are created.

  - To view the state of the newly created OSDs:

    a. Click **Workloads → Pods** from the OpenShift Web Console.

    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

- To view the state of the PVCs:
  a. Click **Storage → Persistent Volume Claims** from the OpenShift Web Console.
  b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

- Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSD devices are encrypted.
  a. Identify the nodes where the new OSD pods are running.

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
```

<OSD-pod-name>

Is the name of the OSD pod.
For example:

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
```

Example output:

```
NODE
compute-1
```

  b. For each of the nodes identified in the previous step, do the following:
    i. Create a debug pod and open a chroot environment for the selected host(s).

```
$ oc debug node/<node-name>
```

<node-name>

Is the name of the node.

```
$ chroot /host
```

    ii. Check for the **crypt** keyword beside the **ocs-deviceset** names.

```
$ lsblk
```



IMPORTANT

Cluster reduction is supported only with the Red Hat Support Team's assistance.

9.2. SCALING OUT STORAGE CAPACITY ON A IBM Z OR IBM LINUXONE CLUSTER

9.2.1. Adding a node using a local storage device

You can add nodes to increase the storage capacity when existing worker nodes are already running at their maximum supported OSDs or when there are not enough resources to add new OSDs on the existing nodes.

Add nodes in the multiple of 3, each of them in different failure domains. Though it is recommended to add nodes in multiples of 3 nodes, you have the flexibility to add one node at a time in flexible scaling deployment. See Knowledgebase article Verify if flexible scaling is enabled



NOTE

OpenShift Data Foundation does not support heterogeneous disk size and types. The new nodes to be added should have the disk of the same type and size which was used during initial OpenShift Data Foundation deployment.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Depending on the type of infrastructure, perform the following steps:
   a. Get a new machine with the required infrastructure. See Platform requirements.
   b. Create a new OpenShift Container Platform worker node using the new machine.
2. Check for certificate signing requests (CSRs) that are in **Pending** state.

   ```
   $ oc get csr
   ```

3. Approve all the required CSRs for the new node.

   ```
   $ oc adm certificate approve <Certificate_Name>
   ```

   <Certificate_Name>

   Is the name of the CSR.

4. Click **Compute → Nodes**, confirm if the new node is in **Ready** state.
5. Apply the OpenShift Data Foundation label to the new node using any one of the following:

   From User interface

a. For the new node, click **Action Menu (⋮) → Edit Labels**.

b. Add **cluster.ocs.openshift.io/openshift-storage**, and click **Save**.

From Command line interface

- Apply the OpenShift Data Foundation label to the new node.

```
$ oc label node <new_node_name> cluster.ocs.openshift.io/openshift-storage=""
```

<new_node_name>

Is the name of the new node.

6. Click **Ecosystem → Installed Operators** from the OpenShift Web Console.
   From the **Project** drop-down list, make sure to select the project where the Local Storage Operator is installed.

7. Click **Local Storage**.

8. Click the **Local Volume Discovery** tab.

   a. Beside the **LocalVolumeDiscovery**, click Action menu **(⋮) → Edit Local Volume Discovery**.

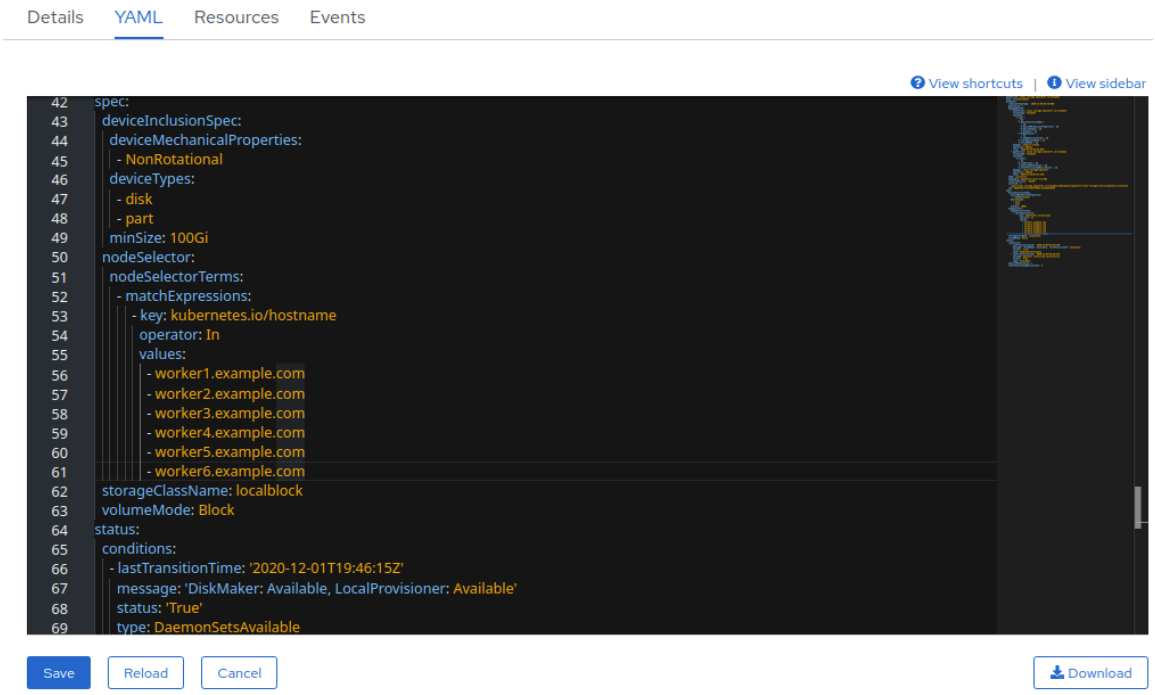
   b. In the YAML, add the hostname of the new node in the **values** field under the node selector.

   c. Click **Save**.

9. Click the **Local Volume Sets** tab.

   a. Beside the **LocalVolumeSet**, click Action menu **(⋮) → Edit Local Volume Set**

   b. In the YAML, add the hostname of the new node in the **values** field under the **node selector**.



YAML showing the addition of new hostnames

c. Click **Save**.



NOTE

It is recommended to add 3 nodes, one each in different zones. You must add 3 nodes and perform this procedure for all of them.

Verification

1. Execute the following command in the terminal and verify that the new node is present in the output:

```
$ oc get nodes --show-labels | grep cluster.ocs.openshift.io/openshift-storage= |cut -d' ' -f1
```

2. On the OpenShift web console, click **Workloads → Pods**, confirm that at least the following pods on the new node are in **Running** state:

- **openshift-storage.cephfs.csi.ceph.com-***
- **openshift-storage.rbd.csi.ceph.com-***

9.2.2. Scaling up storage capacity

To scale up storage capacity, see Scaling up storage capacity on a cluster .

CHAPTER 10. SCALING STORAGE OF IBM POWER OPENSHIFT DATA FOUNDATION CLUSTER

To scale the storage capacity of your configured Red Hat OpenShift Data Foundation worker nodes on IBM Power cluster, you can increase the capacity by adding three disks at a time. Three disks are needed since OpenShift Data Foundation uses a replica count of 3 to maintain the high availability. So the amount of storage consumed is three times the usable space.



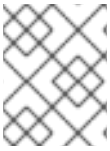
NOTE

Usable space might vary when encryption is enabled or replica 2 pools are being used.

10.1. SCALING UP STORAGE BY ADDING CAPACITY TO YOUR OPENSHIFT DATA FOUNDATION NODES ON IBM POWER INFRASTRUCTURE USING LOCAL STORAGE DEVICES

In order to scale up an OpenShift Data Foundation cluster which was created using local storage devices, a new disk needs to be added to the storage node. It is recommended to have the new disks of the same size as used earlier during the deployment as OpenShift Data Foundation does not support heterogeneous disks/OSD's.

You can add storage capacity (additional storage devices) to your configured local storage based OpenShift Data Foundation worker nodes on IBM Power infrastructures.



NOTE

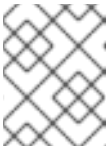
Flexible scaling features get enabled at the time of deployment and can not be enabled or disabled later on.

Prerequisites

- You must be logged into the OpenShift Container Platform cluster.
- You must have installed the local storage operator. Use the following procedure:
  - Installing Local Storage Operator on IBM Power
- You must have three OpenShift Container Platform worker nodes with the same storage type and size attached to each node (for example, 0.5TB SSD) as the original OpenShift Data Foundation StorageCluster was created with.

Procedure

1. To add storage capacity to OpenShift Container Platform nodes with OpenShift Data Foundation installed, you need to
   a. Find the available devices that you want to add, that is, a minimum of one device per worker node. You can follow the procedure for finding available storage devices in the respective deployment guide.



NOTE

Make sure you perform this process for all the existing nodes (minimum of 3) for which you want to add storage.

b. Add the additional disks to the **LocalVolume** custom resource (CR).

```
$ oc edit -n openshift-local-storage localvolume localblock
```

Example output:

```
spec:
  logLevel: Normal
  managementState: Managed
  nodeSelector:
    nodeSelectorTerms:
      - matchExpressions:
          - key: kubernetes.io/hostname
            operator: In
            values:
              - worker-0
              - worker-1
              - worker-2
  storageClassDevices:
    - devicePaths:
        - /dev/sda
        - /dev/sdx    # newly added device
      storageClassName: localblock
      volumeMode: Block
```

Make sure to save the changes after editing the CR.

Example output:

```
localvolume.local.storage.openshift.io/localblock edited
```

You can see in this CR that new devices are added.

- **sdx**

2. Display the newly created Persistent Volumes (PVs) with the **storageclass** name used in the **localVolume** CR.

```
$ oc get pv | grep localblock | grep Available
```

Example output:

```
local-pv-a04ffd8    500Gi    RWO    Delete    Available    localblock    24s
local-pv-a0ca996b    500Gi    RWO    Delete    Available    localblock    23s
local-pv-c171754a    500Gi    RWO    Delete    Available    localblock    23s
```

3. Navigate to the OpenShift Web Console.

4. Click **Storage → Storage cluster**.

a. Click **Actions** at the top right of the page to extend the options menu.

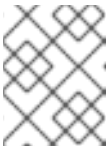
b. Select **Add Capacity** from the options menu.
From this dialog box, set the **Storage Class** name to the name used in the **localVolume** CR.
Available Capacity displayed is based on the local disks available in storage class.

c. Click **Add**.

1. To check the status, navigate to **Storage → Storage cluster** and verify that the **Storage Cluster** in the **Status** card has a green tick.

Verification

- Verify the available Capacity.
  - In the OpenShift Web Console, click **Storage → Storage cluster**.
  - Click the **Block and File** tab, then check the **Raw Capacity** card.
    Note that the capacity increases based on your selections.



NOTE

The raw capacity does not take replication into account and shows the full capacity.

- Verify that the new OSDs and their corresponding new Persistent Volume Claims (PVCs) are created.
  - To view the state of the newly created OSDs:
    a. Click **Workloads → Pods** from the OpenShift Web Console.
    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

  - To view the state of the PVCs:
    a. Click **Storage → Persistent Volume Claims** from the OpenShift Web Console.
    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

- Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSD devices are encrypted.
  a. Identify the nodes where the new OSD pods are running.

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
```

<OSD-pod-name>

Is the name of the OSD pod.
For example:

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
```

Example output:

```
NODE
compute-1
```

b. For each of the nodes identified in the previous step, do the following:

i. Create a debug pod and open a chroot environment for the selected host(s).

```
$ oc debug node/<node-name>
```

<node-name>

Is the name of the node.

```
$ chroot /host
```

ii. Check for the **crypt** keyword beside the **ocs-deviceset** names.

```
$ lsblk
```



IMPORTANT

Cluster reduction is supported only with the Red Hat Support Team's assistance.

10.2. SCALING OUT STORAGE CAPACITY ON A IBM POWER CLUSTER

OpenShift Data Foundation is highly scalable. It can be scaled out by adding new nodes with required storage and enough hardware resources in terms of CPU and RAM. Practically there is no limit on the number of nodes which can be added but from the support perspective 2000 nodes is the limit for OpenShift Data Foundation.

Scaling out storage capacity can be broken down into two steps:

- Adding new node
- Scaling up the storage capacity



NOTE

OpenShift Data Foundation does not support heterogeneous OSD/Disk sizes.

10.2.1. Adding a node using a local storage device on IBM Power

You can add nodes to increase the storage capacity when existing worker nodes are already running at their maximum supported OSDs or when there are not enough resources to add new OSDs on the existing nodes.

Add nodes in the multiple of 3, each of them in different failure domains. Though it is recommended to add nodes in multiples of 3 nodes, you have the flexibility to add one node at a time in flexible scaling deployment. See Knowledgebase article Verify if flexible scaling is enabled



NOTE

OpenShift Data Foundation does not support heterogeneous disk size and types. The new nodes to be added should have the disk of the same type and size which was used during initial OpenShift Data Foundation deployment.

Prerequisites

- You must be logged into the OpenShift Container Platform cluster.
- You must have three OpenShift Container Platform worker nodes with the same storage type and size attached to each node (for example, 2TB SSD drive) as the original OpenShift Data Foundation StorageCluster was created with.

Procedure

1. Get a new IBM Power machine with the required infrastructure. See Platform requirements.
2. Create a new OpenShift Container Platform node using the new IBM Power machine.
   a. Check for certificate signing requests (CSRs) that are in **Pending** state.

      ```
      $ oc get csr
      ```

   b. Approve all the required CSRs for the new node.

      ```
      $ oc adm certificate approve <Certificate_Name>
      ```

      <Certificate_Name>

      Is the name of the CSR.

   c. Click **Compute → Nodes**, confirm if the new node is in **Ready** state.

   d. Apply the OpenShift Data Foundation label to the new node using any one of the following:

      From User interface

      i. For the new node, click **Action Menu (⋮) → Edit Labels**.

      ii. Add **cluster.ocs.openshift.io/openshift-storage** and click **Save**.

      From Command line interface

      - Apply the OpenShift Data Foundation label to the new node.

```
$ oc label node <new_node_name> cluster.ocs.openshift.io/openshift-storage=""
```

<new_node_name>

Is the name of the new node.

e. Click **Ecosystem → Installed Operators** from the OpenShift Web Console.
From the **Project** drop-down list, make sure to select the project where the Local Storage Operator is installed.

f. Click **Local Storage**.

g. Click the **Local Volume** tab.

i. Beside the **LocalVolume**, click Action menu **(⋮) → Edit Local Volume**.

ii. In the YAML, add the hostname of the new node in the **values** field under the **node selector**.
YAML showing the addition of new hostnamesimage::images/ocs_yaml-showing-addition-new-hostnames-localvolume-power.png[Screenshot of YAML showing the addition of new hostnames.]

iii. Click **Save**.



NOTE

It is recommended to add 3 nodes, one each in different zones. You must add 3 nodes and perform this procedure for all of them.

Verification

1. Execute the following command in the terminal and verify that the new node is present in the output:

```
$ oc get nodes --show-labels | grep cluster.ocs.openshift.io/openshift-storage= |cut -d' ' -f1
```

2. On the OpenShift web console, click **Workloads → Pods**, confirm that at least the following pods on the new node are in **Running** state:

- **openshift-storage.cephfs.csi.ceph.com-***
- **openshift-storage.rbd.csi.ceph.com-***

10.2.2. Scaling up storage capacity

To scale up storage capacity, see Scaling up storage capacity on a cluster .

CHAPTER 11. SCALING STORAGE CAPACITY OF IBM FLASHSYSTEM CLUSTER

To scale the storage capacity of your configured Red Hat OpenShift Data Foundation worker nodes on a cluster using external IBM FlashSystem, you can increase the capacity by adding three disks at a time. Three disks are needed since OpenShift Data Foundation uses a replica count of 3 to maintain the high availability. So the amount of storage consumed is three times the usable space.



NOTE

Usable space might vary when encryption is enabled or replica 2 pools are being used.

11.1. SCALING UP STORAGE CAPACITY ON A CLUSTER

To increase the storage capacity in a dynamically created storage cluster on an user-provisioned infrastructure, you can add storage capacity and performance to your configured Red Hat OpenShift Data Foundation worker nodes.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.
- The disk should be of the same size and type as used during initial deployment.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Storage → Storage cluster**.
    a. Click **Actions** at the top right of the page to extend the options menu.
    b. Select **Add Capacity** from the options menu.
    c. Select the **Storage Class**. Choose the storage class which you wish to use to provision new storage devices.
    d. Click **Add**.
3. To check the status, navigate to **Storage → Storage cluster** and verify that the **Storage Cluster** in the Status card has a green tick.

Verification

- Verify the **Raw Capacity** card.
    a. In the OpenShift Web Console, click **Storage → Storage cluster**.
    b. In the **Block and File** tab, check the **Raw Capacity** card.
    Note that the capacity increases based on your selections.



NOTE

The raw capacity does not take replication into account and shows the full capacity.

- Verify that the new object storage devices (OSDs) and their corresponding new Persistent Volume Claims (PVCs) are created.
  - To view the state of the newly created OSDs:
    a. Click **Workloads → Pods** from the OpenShift Web Console.
    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

  - To view the state of the PVCs:
    a. Click **Storage → Persistent Volume Claims** from the OpenShift Web Console.
    b. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

- Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSD devices are encrypted.
  a. Identify the nodes where the new OSD pods are running.

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
```

<OSD-pod-name>

Is the name of the OSD pod.
For example:

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
```

Example output:

```
NODE
compute-1
```

  b. For each of the nodes identified in the previous step, do the following:
    i. Create a debug pod and open a chroot environment for the selected hosts.

```
$ oc debug node/<node-name>
```

<node-name>

Is the name of the node.

```
$ chroot /host
```

ii. Check for the **crypt** keyword beside the **ocs-deviceset** names.

```
$ lsblk
```



IMPORTANT

Cluster reduction is supported only with the Red Hat Support Team's assistance.

11.2. SCALING OUT STORAGE CAPACITY ON AN IBM FLASHSYSTEM CLUSTER

OpenShift Data Foundation is highly scalable. It can be scaled out by adding new nodes with required storage and enough hardware resources in terms of CPU and RAM. Practically there is no limit on the number of nodes which can be added but from the support perspective 2000 nodes is the limit for OpenShift Data Foundation.

Scaling out storage capacity can be broken down into two steps

- Adding new node
- Scaling up the storage capacity



NOTE

OpenShift Data Foundation does not support heterogeneous OSD/Disk sizes.

11.2.1. Adding a node

You can add nodes to increase the storage capacity when existing worker nodes are already running at their maximum supported OSDs or there are not enough resources to add new OSDs on the existing nodes. It is always recommended to add nodes in the multiple of three, each of them in different failure domains.

While we recommend adding nodes in the multiple of three, you still get the flexibility of adding one node at a time in the flexible scaling deployment. Refer to the Knowledgebase article Verify if flexible scaling is enabled.



NOTE

OpenShift Data Foundation does not support heterogeneous disk size and types. The new nodes to be added should have the disk of the same type and size which was used during OpenShift Data Foundation deployment.

11.2.1.1. Adding a node to an installer-provisioned infrastructure

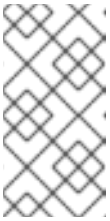
Use this procedure to add a new node to an installer-provisioned infrastructure cluster.

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Navigate to **Compute → Machine Sets**.
2. On the machine set where you want to add nodes, select **Edit Machine Count**
   a. Add the amount of nodes, and click **Save**.
   b. Click **Compute → Nodes** and confirm if the new node is in **Ready** state.
3. Apply the OpenShift Data Foundation label to the new node.
   a. For the new node, click **Action menu (⋮) → Edit Labels**.
   b. Add **cluster.ocs.openshift.io/openshift-storage**, and click **Save**.



NOTE

It is recommended to add 3 nodes, one each in different zones. You must add 3 nodes and perform this procedure for all of them. In case of bare metal installer-provisioned infrastructure deployment, you must expand the cluster first. For instructions, see Expanding the cluster.

Verification

1. Execute the following command in the terminal and verify that the new node is present in the output:

```
$ oc get nodes --show-labels | grep cluster.ocs.openshift.io/openshift-storage= |cut -d' ' -f1
```

2. On the OpenShift web console, click **Workloads → Pods**, confirm that at least the following pods on the new node are in **Running** state:
   - **openshift-storage.cephfs.csi.ceph.com-***
   - **openshift-storage.rbd.csi.ceph.com-***

11.2.1.2. Adding a node to an user-provisioned infrastructure

Prerequisites

- You have administrative privilege to the OpenShift Container Platform Console.
- You have a running OpenShift Data Foundation Storage Cluster.

Procedure

1. Depending on the type of infrastructure, perform the following steps:
   a. Get a new machine with the required infrastructure. See Platform requirements.
   b. Create a new OpenShift Container Platform worker node using the new machine.
2. Check for certificate signing requests (CSRs) that are in **Pending** state.

```
$ oc get csr
```

3. Approve all the required CSRs for the new node.

```
$ oc adm certificate approve <Certificate_Name>
```

<Certificate_Name>
Is the name of the CSR.

4. Click **Compute → Nodes**, confirm if the new node is in **Ready** state.
5. Apply the OpenShift Data Foundation label to the new node using any one of the following:

From User interface

a. For the new node, click **Action Menu (⋮) → Edit Labels**.
b. Add **cluster.ocs.openshift.io/openshift-storage**, and click **Save**.

From Command line interface

- Apply the OpenShift Data Foundation label to the new node.

```
$ oc label node <new_node_name> cluster.ocs.openshift.io/openshift-storage=""
```

<new_node_name>
Is the name of the new node.

Verification

1. Execute the following command in the terminal and verify that the new node is present in the output:

```
$ oc get nodes --show-labels | grep cluster.ocs.openshift.io/openshift-storage= |cut -d' ' -f1
```

2. On the OpenShift web console, click **Workloads → Pods**, confirm that at least the following pods on the new node are in **Running** state:
   - **openshift-storage.cephfs.csi.ceph.com-***
   - **openshift-storage.rbd.csi.ceph.com-***

11.2.2. Scaling up storage capacity

To scale up storage capacity, see Scaling up a cluster.